MRIxFields2026: A Generalizable Cross- Field MRI Translation and Harmonization Challenge: Structured description of the challenge design

CHALLENGE ORGANIZATION

Title

Use the title to convey the essential information on the challenge mission.

MRIxFields2026: A Generalizable Cross- Field MRI Translation and Harmonization Challenge

Challenge acronym

Preferable, provide a short acronym of the challenge (if any).

Magnetic Resonance Imaging × Magnetic Fields 2026, where "×" denotes Cross-Field, highlighting the integration of multi-field MRI acquisitions (from ultra-low to ultra-high field), and data-driven modeling to enable robust image translation, harmonization, and field-aware synthesis across heterogeneous magnetic field strengths.

Challenge abstract

Provide a summary of the challenge purpose. This should include a general introduction in the topic from both a biomedical as well as from a technical point of view and clearly state the envisioned technical and/or biomedical impact of the challenge.

Magnetic resonance imaging (MRI) plays a central role in neuroscience research and clinical assessment of brain structure and function. However, the rapidly expanding spectrum of MRI field strengths, from 0.1T to 7T scanners, has introduced substantial heterogeneity in signal-to-noise ratio, contrast behavior, spatial resolution, and B0/B1 field uniformity. These field-dependent differences fundamentally limit data comparability across centers and pose a major barrier to the generalization and clinical translation of AI-based MRI models.

Despite growing interest in cross-field MRI synthesis and enhancement, the field lacks a standardized, large-scale benchmark that spans the full range of magnetic field strengths and enables systematic evaluation of model robustness, structural fidelity, and field-awareness. Existing datasets and challenges are typically confined to narrow field ranges or single-domain settings, making it difficult to assess whether generative models can generalize across realistic multi-field and multi-center scenarios.

This challenge introduces the first comprehensive public benchmark for multi-field brain MRI image generation, covering field strengths from 0.1T to 7T and incorporating two major structural MRI contrasts (T1w,T2w, and T2-FLAIR). The dataset comprises approximately 500 retrospective cases and 200 prospectively acquired cross-field paired scans, with unified acquisition protocols, standardized preprocessing, and rigorous quality assurance. The paired multi-field acquisitions provide a unique reference for evaluating structural consistency across field transformations.

Building on this dataset, the challenge defines three complementary and clinically motivated tasks:

(1)Ultra-High Field MRI Synthesis from Arbitrary Magnetic Field Strengths
The first task targets the generation of high-field-equivalent MRI from arbitrary input field strengths, enabling models to recover fine anatomical details and quantitative properties associated with 7T imaging. This capability is essential for research centers and early-stage clinical studies where ultrahigh-field scanners are scarce but high-quality representations are indispensable.

(2)Higher-Field MRI Generation from Ultra-Low Magnetic Field Strengths
The second task addresses a rapidly emerging global priority: enhancing ultra-low-field MRI to restore clinically meaningful tissue contrast under severely degraded imaging conditions. As 0.1T–0.3T systems become critical tools for low-resource settings, emergency medicine, and point-of-care imaging, AI-based enhancement becomes a necessary bridge to achieve diagnostic reliability without expensive hardware.

(3)Controllable Field-to-Field MRI Synthesis with a Unified Conditional Model
The third task introduces controllable, generalizable field-to-field synthesis via explicit conditioning mechanisms. This unified modeling strategy is crucial for harmonizing datasets across hospitals, reducing domain shift in multi-center studies, supporting federated learning environments, and enabling large-scale clinical AI deployment across heterogeneous scanners.

Evaluation combines voxel-level accuracy (nRMSE), structural similarity (SSIM), perceptual plausibility (LPIPS), and anatomically grounded segmentation metrics, including mean Dice overlap and normalized volume consistency across 14 bilateral deep gray matter nuclei derived from paired travelling-volunteer scans. These five complementary metrics jointly define the primary leaderboard via rank-summation, enabling balanced assessment of numerical fidelity, perceptual realism, and regional anatomical preservation. Top-ranked submissions will additionally undergo blinded neuroradiologist review to identify hallucinated anatomy or implausible structural alterations.

Together, these tasks establish a unified evaluation framework that moves beyond isolated domain translation and directly tests field-aware and controllable MRI generation.
The challenge aims to advance both methodological and biomedical research by providing a rigorous testbed for assessing robustness, generalizability, and anatomical fidelity of generative models in heterogeneous multi-field MRI environments. By bridging ultra-low-field accessibility and ultra-high-field image quality within a single benchmark, this challenge is expected to catalyze the development of scalable, field-aware MRI synthesis methods and to support reliable multi-center neuroscience research and real-world clinical deployment of low-field MRI systems. The current release establishes a foundational harmonization benchmark, upon which pathology-aware extensions and clinically oriented downstream tasks will be systematically built in future editions.

Challenge keywords

List the primary keywords that characterize the challenge.

Multi-field MRI, Cross-field synthesis, Image harmonization, Generative modeling

Year

2026

Novelty of the challenge

Briefly describe the novelty of the challenge.

1. A comprehensive multi-field MRI dataset spanning 0.1T to 7T.
This challenge introduces the first publicly available MRI benchmark that systematically covers the entire clinically and scientifically relevant magnetic field spectrum, from ultra-low field to ultra-high field MRI systems. Unlike existing datasets that are restricted to narrow field ranges (typically 1.5T–3T or 3T–7T), the proposed dataset integrates ultra-low-field (0.1T), high-field (1.5T and 3T), and ultra-high-field MRI (5T and 7T) within a unified framework.Crucially, the data are acquired from multiple centers and multiple scanner platforms, reflecting realistic inter-site and inter-device variability rather than controlled single-site conditions. This naturally occuring distribution shift enables rigorous evaluation of model robustness, cross-domain generalization, and field-awareness, which are critical yet underexplored challenges in current MRI generative modeling research. In addition, the inclusion of 40 prospectively acquired cross-field paired scans and 100 cases per magnetic field strength (five scanners) provides a rare foundation for directly assessing anatomical consistency across field strengths—an evaluation scenario that is largely absent from existing benchmarks. Importantly, the paired travelling-volunteer acquisitions also enable anatomically grounded evaluation using segmentation-based measures, including Dice overlap and normalized volume consistency across deep gray matter nuclei, providing direct validation of regional structural preservation beyond image-level consistency.

2. A unified task framework for cross-field generation, ultra-low-field enhancement, and controllable field-to-field synthesis.
This challenge proposes a unified benchmark with three tasks that evaluates MRI generation models under three complementary and practically motivated scenarios: 1) high-field (7T-equivalent) synthesis from arbitrary input field strengths, 2) enhancement of severely degraded 0.1T ultra-low-field MRI toward higher-field quality, and 3) controllable any-to-any synthesis between arbitrary field strengths using explicit field conditioning. Rather than treating these problems as isolated tasks, the challenge establishes the first systematic benchmark that jointly assesses any-to-any field transformation and explicit field controllability within a unified dataset and evaluation protocol. This design reflects emerging trends in conditional control, foundation models, and prompt-driven image synthesis, while remaining grounded in clinically meaningful use cases. By requiring models to encode and manipulate field-dependent imaging characteristics, the challenge directly targets the development of field-aware and scalable MRI harmonization methods suitable for real-world deployment.

Task description and application scenarios

Briefly describe the application scenarios for the tasks in the challenge.

Tasks:
(1)Ultra-High Field MRI Synthesis from Arbitrary Magnetic Field Strengths
The first task targets the generation of high-field-equivalent MRI from arbitrary input field strengths, enabling models to recover fine anatomical details and quantitative properties associated with 7T imaging. This capability is essential for research centers and early-stage clinical studies where ultrahigh-field scanners are scarce but high-quality representations are indispensable.

(2)Higher-Field MRI Generation from Ultra-Low Magnetic Field Strengths

The second task addresses a rapidly emerging global priority: enhancing ultra-low-field MRI to restore clinically meaningful tissue contrast under severely degraded imaging conditions. As 0.1T–0.3T systems become critical tools for low-resource settings, emergency medicine, and point-of-care imaging, AI-based enhancement provides a critical pathway to achieve diagnostic reliability without expensive hardware.

(3)Controllable Field-to-Field MRI Synthesis with a Unified Conditional Model
The third task introduces controllable, generalizable field-to-field synthesis via explicit conditioning mechanisms. This unified modeling strategy is crucial for harmonizing datasets across hospitals, reducing domain shift in multi-center studies, supporting federated learning environments, and enabling large-scale clinical AI deployment across heterogeneous scanners.

Applications:
1.Advancing cross-field MRI harmonization and high-field-equivalent image analysis for neuroscience research. The challenge provides a unified dataset spanning 0.1T to 7T, enabling the development of algorithms that mitigate variability introduced by different scanners and magnetic field strengths. Such harmonization is a critical prerequisite for reliable multi-center research, large-scale neuroscience studies, and longitudinal analyses, where inconsistent image quality and field-dependent contrast variations can otherwise confound biological interpretation. By explicitly benchmarking cross-field generation and harmonization within a single evaluation framework, the challenge enables objective assessment of whether generative models can produce structurally consistent and field-invariant representations, including high-field-equivalent images that support downstream regional morphometric consistency and cross-site standardization.

2.Improving the clinical utility and real-world deployability of ultra-low-field MRI systems. Ultra-low-field MRI, including portable 0.1T systems hold significant promise for community healthcare, bedside imaging, emergency medicine, and scenarios involving limited mobility, yet their broader adoption is hindered by reduced signal-to-noise ratio, limited tissue contrast, and susceptibility to artifacts. By promoting algorithms that enhance severely degraded ultra-low-field inputs to higher-field quality, the challenge targets clinically realistic deployment scenarios rather than idealized reconstruction settings. The generated benchmark also provides a rigorous testbed for assessing robustness, failure models, and cross-domain generalizability of AI models under realistic distribution shifts, which are essential criteria for the safe translation of low-field MRI enhancement methods into real-world clinical practice.

FURTHER INFORMATION FOR CONFERENCE ORGANIZERS

Workshop

If the challenge is part of a workshop, please indicate the workshop.

N/A

Duration

How long does the challenge take?

Quarter day.

In case you selected half or full day, please explain why you need a long slot for your challenge.

N/A

Expected number of participants

Please explain the basis of your estimate (e.g. numbers from previous challenges) and/or provide a list of potential participants and indicate if they have already confirmed their willingness to contribute.

We expect approximately 60–100 participating teams worldwide to take part in the MRIxFields2026 challenge. This estimate is based on both empirical evidence from previous related challenges and the broad and timely scope of the proposed benchmark. In particular, recent MRI reconstruction and synthesis challenges that address domain shift, multi-site data, or generative modeling—such as large-scale MRI reconstruction or image translation challenges at MICCAI and ISMRM—have typically attracted 50–80 teams, with strong participation from both academic and industrial research groups. Given the rapidly growing interest in generative foundation models, controllable image synthesis, and low-field MRI enhancement, we anticipate a comparable or higher level of engagement.

Several factors are expected to further increase participation in MRIxFields2026:
1.Unique dataset coverage and novelty: The challenge provides the first public multi-center MRI dataset spanning the full magnetic field range from 0.1T to 7T, addressing an unmet need in the community and appealing to researchers in MRI physics, medical image analysis, and generative AI.

2.Broad methodological relevance: The unified task design (cross-field generation, ultra-low-field enhancement, and controllable field-to-field synthesis) is highly relevant to a wide range of approaches, including diffusion models, transformer-based architectures, conditional and prompt-driven models, and foundation models.

3.Interdisciplinary appeal: The challenge is expected to attract participants from multiple communities, including medical image computing, MRI physics, low-field MRI development, and industry groups working on scalable imaging AI solutions.

Publication and future plans

Please indicate if you plan to coordinate a publication of the challenge results.

Individual challenge papers will be published in LNCS proceedings. After the challenge, we will summarize the results in a challenge paper and submit it to a high-impact journal (e.g., Medical Image Analysis, IEEE Transactions on Medical Imaging).

MICCAI LNCS proceedings

Indicate if you want to offer MICCAI Springer LNCS proceedings to the participants. Publishing a proceedings volume is optional and at the discretion of each challenge's organizers. At a minimum, organizers must ensure that a description of each participant's submission is publicly available. Organizers who wish to publish MICCAI Springer LNCS proceedings must adhere to the MICCAI Satellite events publication process.

Yes

Collaboration with European Society of Radiology (ESR)

In collaboration with European Society of Radiology (ESR), we announce special clinical interest topics with associated clinicians who can help with the preparation of the proposals; the best 3 challenge proposals on these topics will get the opportunity to present their challenges at the European Congress of Radiology (ECR) 2027 in a special session. If you want to organize a challenge in collaboration with ESR on one of these topics, please reach out to the MICCAI Challenges Team (miccai-challenges-2026@dkfz-heidelberg.de) and we will put you in contact with the corresponding clinician.

Challenge in collaboration with ESR. Ticking 'Yes' implies that the challenge has been prepared in collaboration with the clinical contact point.

No

Space and hardware requirements

Organizers of on-site challenges must provide a fair computing environment for all participants. For instance, algorithms should run on the same computing platform provided to all.

Participants are expected to train their models in local computational environments and submit containerized inference pipelines as Docker images via the Synapse platform (https://www.synapse.org/Synapse:syn72060672). Training and validation datasets will be made available for registered teams to download. During the validation phase, a public leaderboard will be maintained on Synapse to provide continuous performance feedback.

For the final evaluation, all submitted Docker containers will be executed on the organizers’ servers under a standardized hardware and software environment to ensure fair and reproducible comparison across methods. Participants are additionally required to report the computational cost associated with model development and inference, and are encouraged to optionally provide estimates of the carbon footprint of their training procedures.
The computing configuration used for the final testing phase is specified as follows:
Operating System: Linux (RockyOS 9)
CPU: 112 cores at 2.0 GHz
RAM: 64 GB
GPU: NVIDIA A6000 (48 GB VRAM, single GPU)
GPU Driver: Version 550
CUDA: Version 12.4
Runtime Limit: 1 hour per team per task

TASK 1: Ultra-High Field MRI Synthesis from Arbitrary Magnetic Field Strengths

SUMMARY

Abstract

Provide a summary of the challenge purpose. This should include a general introduction in the topic from both a biomedical as well as from a technical point of view and clearly state the envisioned technical and/or biomedical impact of the challenge.

Ultrahigh-field (7T) magnetic resonance imaging has fundamentally expanded the ability to probe human brain anatomy and tissue microstructure, offering improved signal-to-noise ratio, enhanced susceptibility contrast, and greater sensitivity to subtle morphological features. These advantages have enabled more precise cortical mapping, refined subcortical delineation, and improved quantification of field-dependent biomarkers that are difficult or impossible to resolve at lower field strengths. As a result, 7T MRI has become an increasingly important tool in neuroscience research, early disease characterization, and the development of quantitative imaging methodologies.

Despite its scientific value, the widespread use of 7T MRI remains constrained by high installation and maintenance costs, limited availability, and regulatory restrictions in clinical settings. Consequently, many large-scale studies and longitudinal cohorts rely predominantly on data acquired at heterogeneous and often lower field strengths, leading to systematic variability in image contrast, resolution, and quantitative measurements. This field-dependent heterogeneity poses a major challenge for cross-center data integration, retrospective analysis, and the translation of high-field insights into broader research and clinical contexts.

Addressing this gap requires methodological advances that can bridge differences across magnetic field strengths while preserving anatomical fidelity and quantitative consistency. In this context, the ability to infer high-field–equivalent image representations from arbitrary input field strengths offers a promising pathway to extend the scientific benefits of ultrahigh-field MRI beyond its physical deployment. By leveraging cross-field information learned from multi-field datasets, such approaches have the potential to enable more uniform structural analysis, enhance the reuse of data from existing cohorts, and support high-resolution investigations in settings where access to 7T systems is limited. This task is designed within this broader scientific motivation, providing a controlled benchmark to study how well current methods can approximate ultrahigh-field characteristics under realistic, heterogeneous acquisition conditions.

This task focuses on generating 7T-equivalent brain MRI from an arbitrary input field strength, establishing ultra-high-field imaging as a unified reference domain. The aim is to reconstruct images that match the fidelity, contrast properties, and fine anatomical detail characteristic of true 7T acquisitions. Participants may design separate models for different input field strengths to address field-dependent variations in noise, resolution, and contrast behavior. The availability of prospectively acquired paired scans with real 7T references enables direct and quantitative assessment of structural consistency and contrast accuracy across field transformations. By defining a fixed high-field output domain, this task offers a rigorous benchmark for advancing generative methods that support multi-field harmonization and broaden access to ultra-high-field image quality. Evaluation leverages the paired travelling-volunteer dataset and includes voxel-level similarity metrics together with

anatomically grounded segmentation measures, including mean Dice overlap and normalized volume consistency across deep gray matter nuclei, ensuring that synthesized images preserve both global fidelity and regional structural integrity.

Keywords

List the primary keywords that characterize the task.

Brain MRI, Ultra-high-field, Image enhancement, Cross-field synthesis

ORGANIZATION

Organizers

a) Provide information on the organizing team (names and affiliations).

Hao Li, Weirui Cai
Institute of Science and Technology for Brain-inspired Intelligence, Fudan University, China

Chengyan Wang
Human Phenome Institute, Fudan University, China

Yuxiang Dai
Department of Psychiatry and Neuroscience, Charité – Universitätsmedizin Berlin, Gemany

Zhiyong Zhang, Cheng Jin
School of Biomedical Engineering at Shanghai Jiao Tong University, China

Zhang Shi
Department of Radiology, Zhongshan Hospital, Fudan University, Shanghai, China

Tianxing He
Department of Biomedical Engineering, Fudan University, China

Yuntong Lyu
School of Medicine, Fudan University, China

b) Provide information on the primary contact person.

Hao Li, h_li@fudan.edu.cn
Institute of Science and Technology for Brain-inspired Intelligence, Fudan University, China

c) Indicate whether clinicians are part of the organizing team. If yes, describe their role.

Yes.
The organizing team includes clinicians with expertise in neuroradiology, clinical neuroscience, and brain MRI, who play a central role in ensuring the clinical relevance, anatomical validity, and translational value of the challenge. They contribute to the design of acquisition protocols across different field strengths, and oversee quality assurance of retrospective and prospectively acquired scans. Clinicians guide the definition of task objectives from a clinical perspective, particularly in relation to high-field-equivalent image fidelity, ultra-low-field

enhancement for diagnostic usability, and cross-field harmonization in multi-center settings.Through their involvement, the clinical organizers help anchor the benchmark in realistic workflows and ensure that the developed methods can support multi-center neuroscience research and future clinical deployment across heterogeneous MRI systems.

Life cycle type

Define the intended submission cycle of the challenge. Include information on whether/how the challenge will be continued after the challenge has taken place.Not every challenge closes after the submission deadline (one-time event). Sometimes it is possible to submit results after the deadline (open call) or the challenge is repeated with some modifications (repeated event).

Examples:

- One-time event with fixed conference submission deadline
- Open call (challenge opens for new submissions after conference deadline)
- Repeated event with annual fixed conference submission deadline

One-time event with fixed conference submission deadline

Challenge venue and platform

a) Report the event (e.g. conference) that is associated with the challenge (if any).

29th International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI 2026).

b) Report the platform used to run the challenge.

synapse.org

c) Do you agree that the your submission is shared with the platform (e.g., grand-challenge, synapse...) that you indicated?

Please note: 1) this purpose of such sharing is that the challenge chairs and the platform can communicate smoothly, your answer won't impact the review of your proposal; 2) regardless of your response to this question, it is your responsibility to perform all actions required by the platform (e.g. filling their submission request).

Yes

d) Provide the URL for the challenge website (if any).

Website for MRIxFields2026: https://mrixfields.github.io/2026; Synapse platform for MRIxFields2026: https://www.synapse.org/Synapse:syn72060672

Participation policies

a) Define the allowed user interaction of the algorithms assessed. This includes the policy regarding any curation, (pre-)processing and (pre-)training steps.

Fully automatic

b) Define the policy on the usage of training data. The data used to train algorithms may, for example, be restricted to the data provided by the challenge or may also include publicly available data including (open) pre-trained nets. Clarify whether such additional data needs to be publicly available at the time of the challenge launch. Clarify whether adding (private) annotations of the public data is allowed.

It should be restricted to the data provided by the current MRIxFields2026 challenges, under the terms and conditions associated with the data usage.

c) Define the participation policy for members of the organizers' institutes. For example, members of the organizers' institutes may participate in the challenge but are not eligible for awards.

May participate but not eligible for awards and not listed in leaderboard

d) Define the award policy. In particular, provide details with respect to challenge prizes.

The top 5 winners receive monetary awards. We are in active negotiation with the sponsor about the value (approximately $2000 in total).

e) Define the policy for result announcement.

Examples:

- Top 3 performing methods will be announced publicly.
- Participating teams can choose whether the performance results will be made public.

All submissions will be reported in the leaderboard. Participating teams can opt out of publication of their results in the leaderboard.

Prize-winning methods will be announced publicly as part of a scientific session at the MICCAI annual meeting.

f) Define the publication policy. In particular, provide details on ...

- ... who of the participating teams/the participating teams' members qualifies as author
- ... whether the participating teams may publish their own results separately, and (if so)
- ... whether an embargo time is defined (so that challenge organizers can publish a challenge paper first).

Participating teams with a valid submission can nominate their team members as co-authors for the challenge paper. We reserve the right to exclude teams if they break the challenge rules. Participating teams can publish their own results but after a 3-month embargo period.

Submission method

a) Describe the method used for result submission. Preferably, provide a link to the submission instructions.

Examples:

- Docker container on the Synapse platform. Link to submission instructions: <URL>
- Algorithm output was sent to organizers via e-mail. Submission instructions were sent by e-mail.

Docker container will be accepted as submission through the Synapse platform. Submission details will be published at the time point of challenge announcement.

b) Provide information on the possibility for participating teams to evaluate their algorithms before submitting final results. For example, many challenges allow submission of multiple results, and only the last run is officially counted to compute challenge results.

Participating teams are allowed to make 3 formal submissions per task. Only the last run submission is officially counted to rank challenge results. Before the final submission on the test set, participants can test their Docker containers on the validation dataset to avoid submission errors.

Challenge schedule

Provide a timetable for the challenge. Preferably, this should include

- the release date(s) of the training cases (if any)
- the registration date/period
- the release date(s) of the test cases and validation cases (if any)
- the submission date(s)
- associated workshop days (if any)
- the release date(s) of the results

[Apr 01, 2026] website opens for registration, release training and validation images
[May 10, 2026] submission system opens for validation
[Aug 01, 2026] submission system opens for testing
[Sept 10, 2026] registration and docker submission deadline
[Oct 08, 2026] release final results during the MICCAI annual meeting

Ethics approval

Indicate whether ethics approval is necessary for the data. If yes, provide details on the ethics approval, preferably institutional review board, location, date and number of the ethics approval (if applicable). Add the URL or a reference to the document of the ethics approval (if available).

Ethics approval has been obtained. The study protocol “Brain MR Multi-Field & Multi-Contrast Image Enhancement” was approved by the Ethics Committee of Zhongshan Hospital, Fudan University, on February 3, 2026 (valid until February 3, 2027). All participants provided written informed consent, and all data will be de-identified prior to release.

Data usage agreement

Clarify how the data can be used and distributed by the teams that participate in the challenge and by others during and after the challenge. This should include the explicit listing of the license applied.

Examples:

- CC BY (Attribution)
- CC BY-SA (Attribution-ShareAlike)
- CC BY-ND (Attribution-NoDerivs)
- CC BY-NC (Attribution-NonCommercial)
- CC BY-NC-SA (Attribution-NonCommercial-ShareAlike)

- CC BY-NC-ND (Attribution-NonCommercial-NoDerivs)

Please note that the data license should not differ among sources. In case a license has to be changed, it has to be reported to the MICCAI challenges team and changed in the proposal.

CC BY-NC (Attribution-NonCommercial)

Code availability

a) Provide information on the accessibility of the organizers' evaluation software (e.g. code to produce rankings). Preferably, provide a link to the code and add information on the supported platforms.

We will provide the synthesis evaluation code to participants before the challenge officially begins. This code will enable participants to test their algorithms and understand the evaluation process.

b) In an analogous manner, provide information on the accessibility of the participating teams' code.

The participating teams are suggested to provide links to their code on our website for reproducibility study. However, it is not a condition of participation.

Conflicts of interest

Provide information related to conflicts of interest. In particular provide information related to sponsoring/funding of the challenge. Also, state explicitly who had/will have access to the test case labels and when.

We declare no conflicts of interest. Test images will only be accessible to the challenge organizers.

MISSION OF THE CHALLENGE

Field(s) of application

State the main field(s) of application that the participating algorithms target.

Examples:

- Diagnosis
- Education
- Intervention assistance
- Intervention follow-up
- Intervention planning
- Prognosis
- Research
- Screening
- Training
- Cross-phase

Research,Diagnosis

Task category(ies)

State the task category(ies)

Examples:

- Classification
- Detection
- Localization
- Modeling
- Prediction
- Reconstruction
- Registration
- Retrieval
- Segmentation
- Tracking

Reconstruction

Cohorts

We distinguish between the target cohort and the challenge cohort. For example, a challenge could be designed around the task of medical instrument tracking in robotic kidney surgery. While the challenge could be based on ex vivo data obtained from a laparoscopic training environment with porcine organs (challenge cohort), the final biomedical application (i.e. robotic kidney surgery) would be targeted on real patients with certain characteristics defined by inclusion criteria such as restrictions regarding sex or age (target cohort).

a) Describe the target cohort, i.e. the subjects/objects from whom/which the data would be acquired in the final biomedical application.

The target cohort for this challenge consists of individuals undergoing brain MRI examinations across a broad range of clinical and research environments. These include patients who may benefit from improved image quality in settings where high-field MRI is unavailable, as well as individuals undergoing neurological evaluation for conditions in which enhanced structural visibility, contrast fidelity, or multi-field harmonization could support diagnosis or longitudinal assessment. The envisioned applications span accessible imaging in low-resource settings, advanced neuroimaging in ultra-high-field environments, and multi-center studies requiring consistent image quality despite variability in scanner hardware and field strength. The ultimate goal is to enable reliable field-aware image generation and enhancement methods that improve the fidelity and comparability of brain MRI for diverse patient populations.

b) Describe the challenge cohort, i.e. the subject(s)/object(s) from whom/which the challenge data was acquired.

The challenge cohort consists of healthy volunteers who underwent brain MRI at five field strengths (0.1T, 1.5T, 3T, 5T, and 7T). The dataset is organized into two complementary components:
1.Unpaired multi-field imaging cohort (non-overlapping subjects):
A total of 500 independent subject–scanner cases, collected as 100 unique healthy subjects per scanner across the five MRI systems. Each case includes two standard brain MRI contrasts acquired under the corresponding system settings. This cohort captures realistic inter-scanner and field-dependent variability in signal-to-noise ratio,

contrast behavior, and spatial resolution, while avoiding subject-level pairing.

2.Travelling-volunteer imaging cohort (paired cross-field scans):
A dedicated cohort of 40 travelling volunteers were scanned on all five MRI systems, yielding 200 paired subject–scanner cases (40 subjects × 5 field strengths). This component provides within-subject cross-field pairing for benchmarking field-to-field transformation consistency and structural fidelity.

No clinical disease labels are provided. Together, the unpaired and paired cohorts enable evaluation of generative models for high-field synthesis, ultra-low-field enhancement, and controllable field-to-field transformation, under both realistic population variability and controlled within-subject comparisons.

Imaging modality(ies)

Specify the imaging technique(s) applied in the challenge.

Magnetic Resonance Imaging

Context information

Provide additional information given along with the images. The information may correspond ...

a) ... directly to the image data (e.g. tumor volume).

None

b) ... to the patient in general (e.g. sex, medical history).

None

Target entity(ies)

a) Describe the data origin, i.e. the region(s)/part(s) of subject(s)/object(s) from whom/which the image data would be acquired in the final biomedical application (e.g. brain shown in computed tomography (CT) data, abdomen shown in laparoscopic video data, operating room shown in video data, thorax shown in fluoroscopy video). If necessary, differentiate between target and challenge cohort.

The data for this challenge originates from brain MRI acquired across a wide range of magnetic field strengths, covering the entire brain including cortical and subcortical regions, deep gray matter structures, and white matter. The imaging focuses exclusively on two commonly used structural MRI contrasts, namely T1-weighted and T2-weighted FLAIR, which are widely adopted in both clinical practice and neuroscience research and exhibit distinct contrast behavior across field strengths.

The challenge cohort consists of healthy volunteers scanned at multiple field strengths between 0.1T and 7T and is composed of two complementary data subsets. The first subset comprises retrospectively collected multi-field cases, acquired from different individuals at different field strengths and therefore non-paired across fields, reflecting realistic variability in acquisition conditions, noise characteristics, contrast behavior, and spatial resolution. The second subset includes prospectively acquired cross-field paired scans, in which the same subjects were imaged at multiple field strengths under controlled protocols. Together, these retrospective and prospective data enable systematic evaluation of generative models for high-field synthesis, ultra-low-field enhancement, and controllable field-to-field transformation across diverse MRI domains.

b) Describe the algorithm target, i.e. the structure(s)/subject(s)/object(s)/component(s) that the participating algorithms have been designed to focus on (e.g. tumor in the brain, tip of a medical instrument, nurse in an operating theater, catheter in a fluoroscopy scan). If necessary, differentiate between target and challenge cohort.

The algorithms in this challenge are designed to generate or enhance brain MRI across different field strengths while preserving anatomical fidelity and contrast-specific tissue characteristics. Participants' models are expected to recover fine anatomical detail, restore tissue contrast, and produce consistent structural representations that align with the appearance of higher-field imaging. Depending on the specific task, the algorithms may generate 7T-equivalent images from arbitrary field strengths, enhance ultra-low-field acquisitions toward higher-field quality, or perform controllable transformations between any pair of field strengths using a unified conditional model. Across all tasks, the goal is to achieve anatomically accurate and contrast-consistent outputs that improve comparability across field strengths and support reliable downstream neuroimaging analyses.

Assessment aim(s)

Identify the property(ies) of the algorithms to be optimized to perform well in the challenge. If multiple properties are assessed, prioritize them (if appropriate). The properties should then be reflected in the metrics applied (see below, parameter metric(s)), and the priorities should be reflected in the ranking when combining multiple metrics that assess different properties.

- Example 1: Find highly accurate liver segmentation algorithm for CT images.
- Example 2: Find lung tumor detection algorithm with high sensitivity and specificity for mammography images.

Corresponding metrics are listed below (parameter metric(s)).

Algorithms should maintain the visual fidelity and structural integrity of reconstructed magnitude images, supporting clinical interpretation. Key metrics include normalized root mean square error (nRMSE), which quantifies intensity differences, and structural similarity index measure (SSIM) , learned perceptual image patch similarity (LPIPS), Dice overlap, and normalized volume consistency, which evaluates structural and detail-level consistency.

DATA SETS

Data source(s)

a) Specify the device(s) used to acquire the challenge data. This includes details on the device(s) used to acquire the imaging data (e.g. manufacturer) as well as information on additional devices used for performance assessment (e.g. tracking system used in a surgical setting).

The challenge data were acquired on five MRI scanners covering field strengths from 0.1T to 7T. These systems cover ultra-low-field, clinical-field, and ultra-high-field imaging environments.
0.1T: piMR-820H (Point Imaging, China)
1.5T: uMR 670 (United Imaging Healthcare, China)
3T: MAGNETOM Prisma (Siemens Healthineers, Germany)
5T: uMR Jupiter (United Imaging Healthcare, China)
7T: MAGNETOM Terra (Siemens Healthineers, Germany)
Raw imaging data include raw and BIDS compatible files, in alignment with the documentation provided in the ethical protocol.

b) Describe relevant details on the imaging process/data acquisition for each acquisition device (e.g. image acquisition protocol(s)).

Imaging protocols include two major brain MRI contrasts: 3D T1-weighted and 2D T2-weighted FLAIR, acquired across multiple field strengths. All data were processed using a unified pipeline consisting of spatial registration, expert-based quality control, and conservative intensity standardization, in accordance with the approved ethical protocol.

Spatial registration. A cascaded registration strategy implemented in ANTs was used to ensure consistent anatomical alignment across subjects, contrasts, and field strengths. All non-3T images were first aligned to a common 3T T1-weighted reference (Siemens MAGNETOM Prisma) using 3D affine registration. The 3T reference image was subsequently registered to the MNI152 T1 1 mm skull-on template using nonlinear SyN registration, and the resulting transforms were composed and applied in a single resampling step to obtain images directly in MNI space.

For multi-contrast data, nonlinear deformation fields were estimated only from T1-weighted images. Corresponding T2-weighted FLAIR images were first aligned to the subject's T1 image using rigid or affine registration, after which the T1-derived deformation fields were applied, ensuring anatomically consistent normalization across contrasts while avoiding modality-driven overfitting. Linear interpolation was used during resampling.

All images underwent expert quality control to exclude scans with severe artifacts or incomplete coverage. Intensity standardization was applied conservatively to reduce scanner-dependent scaling differences while preserving field-strength-specific contrast characteristics. All datasets are organized according to the BIDS specification with consistent metadata across centers.

c) Specify the center(s)/institute(s) in which the data was acquired and/or the data providing platform/source (e.g. previous challenge). If this information is not provided (e.g. for anonymization reasons), specify why.

The following centers have already contacted us and can support our data collection. Additionally, we are actively reaching out to more international medical centers in the hope of collecting raw data from more sources.

Fudan University Brain Imaging Center, Fudan University, China

Point Imaging, Shanghai, China

Sheshan Campus, Zhongshan Hospital, Fudan University, China

These centers participate in retrospective and prospective MRI data collection across multiple field strengths.

d) Describe relevant characteristics (e.g. level of expertise) of the subjects (e.g. surgeon)/objects (e.g. robot) involved in the data acquisition process (if any).

Data are acquired with a team of radiographers and clinical advisors as appropriate.

Training and test case characteristics

a) State what is meant by one case in this challenge. A case encompasses all data that is processed to produce one result that is compared to the corresponding reference result (i.e. the desired algorithm output).

Examples:

- Training and test cases both represent a CT image of a human brain. Training cases have a weak annotation (tumor present or not and tumor volume (if any)) while the test cases are annotated with the tumor contour (if any).
- A case refers to all information that is available for one particular patient in a specific study. This information always includes the image information as specified in data source(s) (see above) and may include context information (see above). Both training and test cases are annotated with survival (binary) 5 years after (first) image was taken.

In this challenge, one case is defined as a subject–scanner (field strength) unit, i.e., the complete set of imaging data acquired from one subject on one MRI system (one field strength). Each case contains two structural brain MRI contrasts (3D T1w and 2D T2-FLAIR) acquired in the same session under that system's acquisition settings, together with the associated metadata required for processing (e.g., field strength/scanner identifier and sequence information).

For the unpaired multi-field cohort, each case corresponds to a unique subject scanned on a single field strength (100 subjects per scanner; 500 cases in total). For the travelling-volunteer cohort, the same subject is scanned on multiple field strengths; each subject–field acquisition is still treated as an independent case, while the shared subject identifier enables optional within-subject cross-field pairing for evaluation of field-to-field transformation consistency and structural fidelity.

b) State the total number of training, validation and test cases.

The dataset contains 700 cases in total (one case = one subject scanned on one MRI system), comprising 500 unpaired cases and 200 travelling-volunteer cases.
The official split is:
Training set: 500 cases (unpaired cohort only; 100 unique subjects per scanner across five field strengths)
Validation set: 100 cases (travelling-volunteer cohort; 20 subjects × 5 field strengths)
Test set: 100 cases (travelling-volunteer cohort; 20 held-out subjects × 5 field strengths)

The travelling-volunteer cohort is split at the subject level to prevent identity leakage between validation and test. No travelling-volunteer data are included in training, ensuring that model development relies on unpaired multi-field data while evaluation is performed under within-subject paired cross-field comparisons.

c) How much of the data are already annotated (stratified by train test in percentage)?

All retrospective data have already been fully annotated, including both training and test sets, while the prospective data are still being collected.

d) Explain why a total number of cases and the specific proportion of training, validation and test cases was chosen.

The total of 700 cases was chosen to jointly capture (i) realistic between-subject and between-scanner variability across field strengths and (ii) rigorous within-subject paired comparisons for cross-field transformation evaluation. Specifically, the 500-case unpaired cohort (100 unique subjects per scanner across five field strengths) provides sufficient diversity in anatomy, acquisition conditions, and field-dependent image characteristics to support robust model training without relying on paired supervision.

The travelling-volunteer cohort was reserved exclusively for evaluation and split evenly into validation (100 cases; 20 subjects × 5 fields) and test (100 cases; 20 subjects × 5 fields). This design enables statistically stable model selection and final ranking using within-subject cross-field pairing, while enforcing subject-level holdout to prevent identity leakage. By keeping all paired travelling data out of training, the split preserves the intended challenge focus on learning cross-field mappings from unpaired multi-field data, while still providing a controlled and fair benchmark for assessing transformation consistency and structural fidelity across field strengths.

e) Mention further important characteristics of the training, validation and test cases (e.g. class distribution in classification tasks chosen according to real-world distribution vs. equal class distribution) and justify the choice.

The unpaired training cohort intentionally reflects a broad real-world spectrum, including subjects with a wide age range and both healthy volunteers and clinical patients, so that models are exposed to realistic anatomical variability and acquisition heterogeneity. In contrast, the travelling-volunteer cohort used for validation and testing consists of young healthy volunteers, because acquiring within-subject paired scans across five field strengths is logistically demanding, costly, and ethically challenging in patient populations. We therefore use the travelling cohort primarily to provide a controlled paired benchmark for assessing cross-field transformation consistency and structural fidelity, while the diverse training cohort supports robustness to population variability encountered in practice.

f) Challenge organizers are encouraged to (partly) use unseen, unpublished data for their challenges. Describe if new data will be used for the challenge and state the number of cases along with the proportion of new data.

All data used in this challenge are newly collected and have not been previously published. The entire dataset—including both the retrospective cohort (approximately 500 cases) and the prospectively acquired cross-field paired scans (approximately 200 cases)—has been collected and curated exclusively by the organizing team. As a result, 100% of the cases in this challenge consist of unseen, unpublished data, ensuring a fair evaluation setting and preventing any prior exposure or data leakage.

Annotation characteristics

a) Describe the method for determining the reference annotation, i.e. the desired algorithm output. Provide the information separately for the training, validation and test cases if necessary. Possible methods include manual image annotation, in silico ground truth generation and annotation by automatic methods.

If human annotation was involved, state the number of annotators.

This challenge does not involve manual annotation, as the task focuses on image generation and field-to-field synthesis rather than structural delineation or semantic labeling. The reference outputs are therefore defined entirely by the imaging data themselves.

Training cases (unpaired cohort):
No paired reference outputs are provided. Training data consist of real acquired whole-brain MRI across field strengths (including 7T as the target-domain distribution), enabling unpaired/unsupervised learning for 7T-equivalent synthesis.

Validation and test cases (travelling-volunteer cohort; paired):
The reference output is obtained by direct acquisition: for each travelling subject and each contrast, the true 7T

image acquired for the same subject serves as the reference target for Task 1. This provides a within-subject paired benchmark for quantitative comparison of the synthesized 7T-equivalent outputs.

b) Provide the instructions given to the annotators (if any) prior to the annotation. This may include description of a training phase with the software. Provide the information separately for the training, validation and test cases if necessary. Preferably, provide a link to the annotation protocol.

No instructions are required, as this challenge does not involve human annotators or manual labeling. All reference outputs are obtained from standardized MRI acquisition and preprocessing pipelines. Consistency across cases is ensured through unified acquisition guidelines and harmonized preprocessing following BIDS conventions rather than through manual annotation practices.

c) Provide details on the subject(s)/algorithm(s) that annotated the cases (e.g. information on level of expertise such as number of years of professional experience, medically-trained or not). Provide the information separately for the training, validation and test cases if necessary.

This challenge does not include any human annotators. No segmentation, contouring, or manual interpretation is used to generate reference outputs for training, validation, or test datasets. All reference images are derived directly from the corresponding MRI acquisitions.

d) Describe the method(s) used to merge multiple annotations for one case (if any). Provide the information separately for the training, validation and test cases if necessary.

Not applicable. Since no manual annotations are generated and no annotators are involved, there are no multiple annotation sources to merge. All reference outputs are obtained from single-source imaging data collected under standardized acquisition protocols.

Data pre-processing method(s)

Describe the method(s) used for pre-processing the raw training data before it is provided to the participating teams. Provide the information separately for the training, validation and test cases if necessary.

N/A

Sources of error

a) Describe the most relevant possible error sources related to the image annotation. If possible, estimate the magnitude (range) of these errors, using inter-and intra-annotator variability, for example. Provide the information separately for the training, validation and test cases, if necessary.

N.A.

b) In an analogous manner, describe and quantify other relevant sources of error.

N.A.

ASSESSMENT METHODS

Metric(s)

a) Define the metric(s) to assess a property of an algorithm. These metrics should reflect the desired algorithm properties described in assessment aim(s) (see above). State which metric(s) were used to compute the ranking(s) (if any).

- Example 1: Dice Similarity Coefficient (DSC)
- Example 2: Area under curve (AUC)

The assessment methodology for Task 1 focuses on evaluating image fidelity, structural consistency, and perceptual quality in high-field MRI synthesis, where ultra-high-field (7T) images serve as the reference domain. To assess the quality of generated 7T-equivalent images from arbitrary input field strengths, the following complementary metrics are employed:
Normalized Root Mean Square Error (nRMSE): nRMSE quantifies global intensity and contrast deviations between the synthesized images and the 7T reference images, providing a measure of numerical reconstruction accuracy in high-field image synthesis.
Structural Similarity Index Measure (SSIM): SSIM evaluates structural and perceptual similarity between generated images and 7T references, with sensitivity to the preservation of anatomical structures and contrast relationships characteristic of ultra-high-field MRI.
Learned Perceptual Image Patch Similarity (LPIPS): LPIPS measures perceptual similarity based on deep feature representations and captures high-level structural and textural differences that are relevant to visual realism and anatomical plausibility in synthesized high-field images.
Dice overlap (deep gray matter): Dice similarity is computed on automated SynthSeg-derived segmentations of 14 bilateral deep gray matter nuclei (thalamus, caudate, putamen, pallidum, hippocampus, amygdala, and accumbens). Dice quantifies spatial overlap and boundary consistency between synthesized images and paired 7T references, providing anatomically grounded assessment of regional structural preservation.

Normalized volume consistency (deep gray matter): For each deep gray matter structure, volume consistency is defined as 1 - |Vsynth - Vref| / Vref, where Vsynth and Vref denote the synthesized and reference regional volumes, respectively. This bounded "higher-is-better" score captures relative volumetric deviation and complements Dice by detecting systematic expansion or contraction not reflected by spatial overlap alone.

Together, these five metrics provide complementary perspectives on numerical accuracy, structural fidelity, perceptual realism, and regional anatomical preservation in 7T-equivalent MRI generation.

b) Justify why the metric(s) was/were chosen, preferably with reference to the biomedical application.

The selected metrics reflect the biomedical objectives of Task 1, which aims to synthesize images that approximate the quality and structural detail of ultra-high-field (7T) MRI. nRMSE ensures accurate reproduction of global signal intensity and contrast characteristics associated with high-field imaging. SSIM evaluates the preservation of anatomical structures, including cortical boundaries, deep gray matter nuclei, and fine spatial detail that are enhanced at 7T. LPIPS complements these measures by assessing perceptual similarity aligned with human visual judgement, providing sensitivity to subtle textural and contrast variations that influence neuroanatomical interpretation. Deep gray matter nuclei are specifically selected because they are anatomically compact, clinically meaningful, and reliably segmentable across field strengths, making them sensitive probes for synthesis-induced distortions such as boundary shifts or volumetric bias.

Combined, these metrics enable a balanced evaluation of numerical accuracy, anatomical integrity, and perceptual quality, supporting rigorous assessment of generative models for high-field MRI synthesis.

Ranking method(s)

a) Describe the method used to compute a performance rank for all submitted algorithms based on the generated metric results on the test cases. Typically the text will describe how results obtained per case and metric are aggregated to arrive at a final score/ranking. Ideally, provide the ranking scheme as a concrete pseudo code.

The final ranking of all submitted algorithms is based on their overall performance across all test cases, as follows:
1. Individual Metric Ranking: For each metric (nRMSE, SSIM, LPIPS,mean Dice overlap, and mean normalized volume consistency), the metric is first computed for every test case and then averaged across all test cases. Lower values are preferred for nRMSE and LPIPS, while higher values are preferred for SSIM,Dice overlap, and normalized volume consistency.
. Each algorithm is ranked independently for each metric based on its mean performance.
2. Overall Ranking: The overall ranking is obtained by summing the per-metric ranks for each algorithm. Algorithms with the lowest composite scores achieve the highest final rank. This rank-summation approach ensures that differences in metric scales do not influence the overall evaluation. This procedure yields a stable and interpretable ranking that reflects performance across complementary dimensions of image quality.

b) Describe the method(s) used to manage submissions with missing results on test cases.

Participating teams are required to submit Docker containers and process all the cases in the test set on our server.Metrics will be computed only on valid cases; however, if an algorithm produces invalid outputs for any test cases, it will be penalized by assigning the worst affected metric, preventing strategic omission of difficult samples.

c) Justify why the described ranking scheme(s) was/were used.

The composite rank-summation scheme is chosen to ensure fair and balanced evaluation across multiple complementary aspects of image quality. nRMSE measures numerical agreement, SSIM captures structural fidelity, and LPIPS reflects perceptual realism,while Dice overlap and normalized volume consistency explicitly assess regional deep gray matter preservation. By ranking algorithms separately for each metric and combining ranks rather than raw metric values, the method avoids biases arising from differences in metric scale or dynamic range.

This multi-metric ranking approach provides a robust and anatomically grounded assessment by jointly considering voxel-level fidelity, perceptual realism, and deep gray matter structural preservation.

In addition, the top-ranked submissions will undergo blinded visual inspection by experienced neuroradiologists to identify hallucinated anatomy or anatomically implausible alterations, serving as a qualitative safeguard beyond automated metrics.

Statistical analyses

Provide an overview of the statistical approaches used in the scope of the challenge analysis. Details can be provided in the parameters below. For each parameter, justify why the described statistical method(s) was/were used and, if necessary, add a description of any method used to assess whether the data met the assumptions required for the particular statistical approach.

The statistical analysis framework of MRIxFields2026 is designed to ensure robust, unbiased, and clinically meaningful evaluation of cross-field MRI generation algorithms. Given the heterogeneous nature of the dataset—spanning magnetic field strengths from 0.1T to 7T, multiple scanners, and both paired and unpaired acquisitions—the challenge adopts case-wise, distribution-aware, and rank-robust statistical methods.

Performance is primarily assessed on held-out, unseen test cases, with all metrics computed at the individual-case level and subsequently aggregated to support population-level inference. Where applicable, non-parametric statistics are preferred to avoid assumptions of normality, and all analyses are conducted under a unified evaluation protocol to ensure comparability across tasks and teams.

Provide a description of how the precision of the performance estimates of individual algorithms is assessed (e.g. confidence interval of the mean on the test set computed using percentile bootstrap, confidence interval of the accuracy on the test set computed using percentile bootstrap).

The precision of performance estimates for each submitted algorithm is quantified using bootstrap-based confidence intervals. Specifically, for each evaluation metric, percentile bootstrap resampling (e.g., 1,000 resamples with replacement at the case level) is applied to the test set to estimate 95% confidence intervals of the mean performance. This approach is well suited to the challenge setting, as it does not rely on distributional assumptions and is robust to skewed or heavy-tailed metric distributions that commonly arise in cross-domain MRI evaluation. Confidence intervals are reported alongside point estimates to reflect uncertainty and to enable fair comparison between methods.

Provide a description of how variability of the performance of individual algorithms across tests cases is assessed (e.g. SD across test cases, IQR, graphs, reporting outliers...).

To characterize how algorithm performance varies across individual test cases, case-wise variability analyses are performed. For each method and metric, variability is summarized using the standard deviation (SD) and interquartile range (IQR) across test cases, providing complementary views of dispersion and robustness. In addition, case-level performance distributions (e.g., boxplots or violin plots) are used to visualize heterogeneity, identify outliers, and highlight failure modes under specific field strengths or acquisition conditions.

Provide a description of how variability of rankings is assessed.

To evaluate the stability of algorithm rankings, variability across test cases is quantified using standard measures such as the standard deviation and interquartile range of per-metric performance. Pairwise rank correlations are computed to assess agreement between rankings derived from nRMSE, SSIM, LPIPS, dice overlap and normalized volume consistency. These analyses ensure that the final composite ranking is robust to minor fluctuations in individual metric values.

Provide a description of statistical tests that are used to assess whether the differences in performance between algorithms are statistically significant.

To assess whether observed performance differences between algorithms are statistically significant, paired, non-parametric hypothesis tests are employed. For pairwise comparisons between methods on the same test cases, the Wilcoxon signed-rank test is used, as it accounts for within-case pairing and does not assume normality. When comparing more than two algorithms simultaneously, a Friedman test is applied, followed by post-hoc pairwise comparisons with multiple-comparison control using the Benjamini–Hochberg false discovery rate (FDR) procedure. Multiple-comparison correction will be applied across all algorithm comparisons within each metric to control the expected false discovery rate.

Provide a description of the missing data handling.

All test cases are verified for completeness prior to release, and missing data are not expected. If a submission fails to produce an output for a specific case, that case will be assigned the lowest possible metric score as defined in the ranking procedure, ensuring consistent treatment across teams.

Indicate any software product that is used for all data analysis methods.

All statistical analyses and visualizations are performed using Python-based tools, including NumPy, SciPy, and Matplotlib. These libraries support rank correlation analysis, variability assessment, and aggregate metric evaluation needed for the challenge.

Further analyses

Present further analyses to be performed (if applicable), e.g. related to

- combining algorithms via ensembling,
- inter-algorithm variability,
- common problems/biases of the submitted methods, or
- ranking variability.

N/A

TASK 2: Higher-Field MRI Generation from Ultra-Low Magnetic Field Strengths

SUMMARY

Abstract

Provide a summary of the challenge purpose. This should include a general introduction in the topic from both a biomedical as well as from a technical point of view and clearly state the envisioned technical and/or biomedical impact of the challenge.

In recent years, ultra-low-field (0.1T) MRI has attracted growing interest due to its low cost, portability, and potential for broader deployment in resource-limited and point-of-care settings. These systems offer unique advantages in accessibility, but their clinical utility is substantially constrained by severely reduced signal-to-noise ratio, limited spatial resolution, and diminished tissue contrast, which impair anatomical visibility and diagnostic interpretability.

Unlike conventional clinical MRI, ultra-low-field imaging operates under fundamentally different signal and noise regimes, where reduced polarization and altered contrast mechanisms lead to pronounced degradation in tissue separability and structural detail. As a result, many anatomical features that are readily visible at clinical field strengths become ambiguous or indistinguishable at ultra-low field. These limitations pose a significant barrier to the broader clinical adoption of ultra-low-field MRI, as conventional reconstruction pipelines and post-processing strategies developed for clinical-field systems often fail to generalize to such severely degraded acquisition conditions. Bridging this gap requires data-driven approaches that can robustly infer higher-field–like anatomical structure and contrast from ultra-low-field inputs, while remaining resilient to extreme noise, resolution loss, and domain shift.

This task focuses on generating higher-field brain MRI from 0.1T ultra-low-field inputs, with the goal of enabling clinically meaningful image quality from an otherwise highly degraded acquisition regime while explicitly preserving clinically relevant anatomical structures that are critical for downstream neuroanatomical interpretation. Rather than simple visual enhancement, the objective is to recover anatomically and diagnostically relevant tissue contrast and structural information that are typically unresolved at ultra-low field strengths. Participants are encouraged to develop generative models capable of operating under extreme noise and resolution limitations, where conventional reconstruction or enhancement approaches are often insufficient.

The availability of paired multi-field acquisitions enables direct and quantitative evaluation of how effectively synthesized images approximate true higher-field references in terms of structural fidelity and contrast restoration. Evaluation includes voxel-level similarity metrics (nRMSE, SSIM, LPIPS) together with anatomically grounded segmentation measures, including mean Dice overlap and normalized volume consistency across deep gray matter nuclei, ensuring that enhancement preserves both global fidelity and regional structural integrity. By focusing on ultra-low-field inputs, this task highlights the potential of generative MRI methods to extend the clinical usability of ultra-low-field systems, while naturally constituting a stringent evaluation scenario for model robustness under pronounced distribution shifts, particularly for extreme field transitions (0.1T to clinical-field strengths). Together with fixed-target high-field synthesis, Task 2 complements the challenge by addressing a value-driven and highly constrained imaging setting where generative modeling can have substantial practical

impact.

Keywords

List the primary keywords that characterize the task.

Ultra-low-field MRI, MRI enhancement, Generative modeling, Cross-field generalization

ORGANIZATION

Organizers

a) Provide information on the organizing team (names and affiliations).

Weirui Cai, Hao Li, He Wang, Hui Zhang, Xueqin Xia, Xin Guo, Chang Xu
Institute of Science and Technology for Brain-inspired Intelligence, Fudan University, China

Fangrong Zong, Yong Liu
School of Artificial Intelligence, Beijing University of Posts and Telecommunications, China

Jiahao Huang
Department of Bioengineering and I-X, Imperial Collegel London, UK

Juergen Hennig
University Medical Center FREIBURG, Germany

Xingfeng Shao
Institute for Medical Imaging Technology, Ruijin Hospital, Shanghai Jiao Tong University
School of Medicine, Shanghai, China

Zixuan Lin
College of Biomidical Engineering & Instrument Science, Zhejiang University, China

Tianxing He
Department of Biomedical Engineering, Fudan University, China

Yuntong Lyu
School of Medicine, Fudan University, China

b) Provide information on the primary contact person.

Weirui Cai, weiruicai@fudan.edu.cn
Institute of Science and Technology for Brain-inspired Intelligence, Fudan University, China

c) Indicate whether clinicians are part of the organizing team. If yes, describe their role.

Yes.
The organizing team includes clinicians with expertise in neuroradiology, clinical neuroscience, and brain MRI, who play a central role in ensuring the clinical relevance, anatomical validity, and translational value of the

challenge. They contribute to the design of acquisition protocols across different field strengths, and oversee quality assurance of retrospective and prospectively acquired scans. Clinicians guide the definition of task objectives from a clinical perspective, particularly in relation to high-field-equivalent image fidelity, ultra-low-field enhancement for diagnostic usability, and cross-field harmonization in multi-center settings.Through their involvement, the clinical organizers help anchor the benchmark in realistic workflows and ensure that the developed methods can support multi-center neuroscience research and future clinical deployment across heterogeneous MRI systems.

Life cycle type

Define the intended submission cycle of the challenge. Include information on whether/how the challenge will be continued after the challenge has taken place.Not every challenge closes after the submission deadline (one-time event). Sometimes it is possible to submit results after the deadline (open call) or the challenge is repeated with some modifications (repeated event).

Examples:

- One-time event with fixed conference submission deadline
- Open call (challenge opens for new submissions after conference deadline)
- Repeated event with annual fixed conference submission deadline

One-time event with fixed conference submission deadline

Challenge venue and platform

a) Report the event (e.g. conference) that is associated with the challenge (if any).

29th International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI 2026).

b) Report the platform used to run the challenge.

synapse.org

c) Do you agree that the your submission is shared with the platform (e.g., grand-challenge, synapse...) that you indicated?

Please note: 1) this purpose of such sharing is that the challenge chairs and the platform can communicate smoothly, your answer won't impact the review of your proposal; 2) regardless of your response to this question, it is your responsibility to perform all actions required by the platform (e.g. filling their submission request).

Yes

d) Provide the URL for the challenge website (if any).

Website for MRIxFields2026: https://mrixfields.github.io/2026; Synapse platform for MRIxFields2026: https://www.synapse.org/Synapse:syn72060672

Participation policies

a) Define the allowed user interaction of the algorithms assessed. This includes the policy regarding any curation, (pre-)processing and (pre-)training steps.

Fully automatic

b) Define the policy on the usage of training data. The data used to train algorithms may, for example, be restricted to the data provided by the challenge or may also include publicly available data including (open) pre-trained nets. Clarify whether such additional data needs to be publicly available at the time of the challenge launch. Clarify whether adding (private) annotations of the public data is allowed.

It should be restricted to the data provided by the current MRIxFields2026 challenges, under the terms and conditions associated with the data usage.

c) Define the participation policy for members of the organizers' institutes. For example, members of the organizers' institutes may participate in the challenge but are not eligible for awards.

May participate but not eligible for awards and not listed in leaderboard

d) Define the award policy. In particular, provide details with respect to challenge prizes.

The top 5 winners receive monetary awards. We are in active negotiation with the sponsor about the value (approximately $2000 in total).

e) Define the policy for result announcement.

Examples:

- Top 3 performing methods will be announced publicly.
- Participating teams can choose whether the performance results will be made public.

All submissions will be reported in the leaderboard. Participating teams can opt out of publication of their results in the leaderboard.

Prize-winning methods will be announced publicly as part of a scientific session at the MICCAI annual meeting.

f) Define the publication policy. In particular, provide details on ...

- ... who of the participating teams/the participating teams’ members qualifies as author
- ... whether the participating teams may publish their own results separately, and (if so)
- ... whether an embargo time is defined (so that challenge organizers can publish a challenge paper first).

Participating teams with a valid submission can nominate their team members as co-authors for the challenge paper. We reserve the right to exclude teams if they break the challenge rules. Participating teams can publish their own results but after a 3-month embargo period.

Submission method

a) Describe the method used for result submission. Preferably, provide a link to the submission instructions.

Examples:

- Docker container on the Synapse platform. Link to submission instructions: <URL>
- Algorithm output was sent to organizers via e-mail. Submission instructions were sent by e-mail.

Docker container will be accepted as submission through the Synapse platform. Submission details will be published at the time point of challenge announcement.

b) Provide information on the possibility for participating teams to evaluate their algorithms before submitting final results. For example, many challenges allow submission of multiple results, and only the last run is officially counted to compute challenge results.

Participating teams are allowed to make 3 formal submissions per task. Only the last run submission is officially counted to rank challenge results. Before the final submission on the test set, participants can test their Docker containers on the validation dataset to avoid submission errors.

Challenge schedule

Provide a timetable for the challenge. Preferably, this should include

- the release date(s) of the training cases (if any)
- the registration date/period
- the release date(s) of the test cases and validation cases (if any)
- the submission date(s)
- associated workshop days (if any)
- the release date(s) of the results

[Apr 01, 2026] website opens for registration, release training and validation images
[May 20, 2026] submission system opens for validation
[Aug 01, 2026] submission system opens for testing
[Sept 10, 2026] registration and docker submission deadline
[Oct 08, 2026] release final results during the MICCAI annual meeting

Ethics approval

Indicate whether ethics approval is necessary for the data. If yes, provide details on the ethics approval, preferably institutional review board, location, date and number of the ethics approval (if applicable). Add the URL or a reference to the document of the ethics approval (if available).

We are currently applying for ethical approval, which is expected to be granted within approximately one month.

Data usage agreement

Clarify how the data can be used and distributed by the teams that participate in the challenge and by others during and after the challenge. This should include the explicit listing of the license applied.

Examples:

- CC BY (Attribution)
- CC BY-SA (Attribution-ShareAlike)
- CC BY-ND (Attribution-NoDerivs)
- CC BY-NC (Attribution-NonCommercial)

- CC BY-NC-SA (Attribution-NonCommercial-ShareAlike)
- CC BY-NC-ND (Attribution-NonCommercial-NoDerivs)

Please note that the data license should not differ among sources. In case a license has to be changed, it has to be reported to the MICCAI challenges team and changed in the proposal.

CC BY-NC (Attribution-NonCommercial)

Code availability

a) Provide information on the accessibility of the organizers' evaluation software (e.g. code to produce rankings). Preferably, provide a link to the code and add information on the supported platforms.

We will provide the synthesis evaluation code to participants before the challenge officially begins. This code will enable participants to test their algorithms and understand the evaluation process.

b) In an analogous manner, provide information on the accessibility of the participating teams' code.

The participating teams are suggested to provide links to their code on our website for reproducibility study. However, it is not a condition of participation.

Conflicts of interest

Provide information related to conflicts of interest. In particular provide information related to sponsoring/funding of the challenge. Also, state explicitly who had/will have access to the test case labels and when.

We declare no conflicts of interest. Test images will only be accessible to the challenge organizers.

MISSION OF THE CHALLENGE

Field(s) of application

State the main field(s) of application that the participating algorithms target.

Examples:

- Diagnosis
- Education
- Intervention assistance
- Intervention follow-up
- Intervention planning
- Prognosis
- Research
- Screening
- Training
- Cross-phase

Research,Diagnosis

Task category(ies)

State the task category(ies)

Examples:

- Classification
- Detection
- Localization
- Modeling
- Prediction
- Reconstruction
- Registration
- Retrieval
- Segmentation
- Tracking

Reconstruction

Cohorts

We distinguish between the target cohort and the challenge cohort. For example, a challenge could be designed around the task of medical instrument tracking in robotic kidney surgery. While the challenge could be based on ex vivo data obtained from a laparoscopic training environment with porcine organs (challenge cohort), the final biomedical application (i.e. robotic kidney surgery) would be targeted on real patients with certain characteristics defined by inclusion criteria such as restrictions regarding sex or age (target cohort).

a) Describe the target cohort, i.e. the subjects/objects from whom/which the data would be acquired in the final biomedical application.

The target cohort for this challenge consists of individuals undergoing brain MRI examinations across a broad range of clinical and research environments. These include patients who may benefit from improved image quality in settings where high-field MRI is unavailable, as well as individuals undergoing neurological evaluation for conditions in which enhanced structural visibility, contrast fidelity, or multi-field harmonization could support diagnosis or longitudinal assessment. The envisioned applications span accessible imaging in low-resource settings, advanced neuroimaging in ultra-high-field environments, and multi-center studies requiring consistent image quality despite variability in scanner hardware and field strength. The ultimate goal is to enable reliable field-aware image generation and enhancement methods that improve the fidelity and comparability of brain MRI for diverse patient populations.

b) Describe the challenge cohort, i.e. the subject(s)/object(s) from whom/which the challenge data was acquired.

The challenge cohort consists of healthy volunteers who underwent brain MRI at five field strengths (0.1T, 1.5T, 3T, 5T, and 7T). The dataset is organized into two complementary components:
1.Unpaired multi-field imaging cohort (non-overlapping subjects):
A total of 500 independent subject–scanner cases, collected as 100 unique healthy subjects per scanner across

the five MRI systems. Each case includes two standard brain MRI contrasts acquired under the corresponding system settings. This cohort captures realistic inter-scanner and field-dependent variability in signal-to-noise ratio, contrast behavior, and spatial resolution, while avoiding subject-level pairing.

2.Travelling-volunteer imaging cohort (paired cross-field scans):
A dedicated cohort of 40 travelling volunteers were scanned on all five MRI systems, yielding 200 paired subject–scanner cases (40 subjects × 5 field strengths). This component provides within-subject cross-field pairing for benchmarking field-to-field transformation consistency and structural fidelity.

No clinical disease labels are provided. Together, the unpaired and paired cohorts enable evaluation of generative models for high-field synthesis, ultra-low-field enhancement, and controllable field-to-field transformation, under both realistic population variability and controlled within-subject comparisons.

Imaging modality(ies)

Specify the imaging technique(s) applied in the challenge.

Magnetic Resonance Imaging

Context information

Provide additional information given along with the images. The information may correspond ...

a) ... directly to the image data (e.g. tumor volume).

None

b) ... to the patient in general (e.g. sex, medical history).

None

Target entity(ies)

a) Describe the data origin, i.e. the region(s)/part(s) of subject(s)/object(s) from whom/which the image data would be acquired in the final biomedical application (e.g. brain shown in computed tomography (CT) data, abdomen shown in laparoscopic video data, operating room shown in video data, thorax shown in fluoroscopy video). If necessary, differentiate between target and challenge cohort.

The data for this challenge originates from brain MRI acquired across a wide range of magnetic field strengths, covering the entire brain including cortical and subcortical regions, deep gray matter structures, and white matter. The imaging focuses exclusively on two commonly used structural MRI contrasts, namely T1-weighted and T2-weighted FLAIR, which are widely adopted in both clinical practice and neuroscience research and exhibit distinct contrast behavior across field strengths.

The challenge cohort consists of healthy volunteers scanned at multiple field strengths between 0.1T and 7T and is composed of two complementary data subsets. The first subset comprises retrospectively collected multi-field cases, acquired from different individuals at different field strengths and therefore non-paired across fields, reflecting realistic variability in acquisition conditions, noise characteristics, contrast behavior, and spatial resolution. The second subset includes prospectively acquired cross-field paired scans, in which the same subjects were imaged at multiple field strengths under controlled protocols. Together, these retrospective and prospective data enable systematic evaluation of generative models for high-field synthesis, ultra-low-field enhancement, and controllable field-to-field transformation across diverse MRI domains.

b) Describe the algorithm target, i.e. the structure(s)/subject(s)/object(s)/component(s) that the participating algorithms have been designed to focus on (e.g. tumor in the brain, tip of a medical instrument, nurse in an operating theater, catheter in a fluoroscopy scan). If necessary, differentiate between target and challenge cohort.

The algorithms in this challenge are designed to generate or enhance brain MRI across different field strengths while preserving anatomical fidelity and contrast-specific tissue characteristics. Participants' models are expected to recover fine anatomical detail, restore tissue contrast, and produce consistent structural representations that align with the appearance of higher-field imaging. Depending on the specific task, the algorithms may generate 7T-equivalent images from arbitrary field strengths, enhance ultra-low-field acquisitions toward higher-field quality, or perform controllable transformations between any pair of field strengths using a unified conditional model. Across all tasks, the goal is to achieve anatomically accurate and contrast-consistent outputs that improve comparability across field strengths and support reliable downstream neuroimaging analyses.

Assessment aim(s)

Identify the property(ies) of the algorithms to be optimized to perform well in the challenge. If multiple properties are assessed, prioritize them (if appropriate). The properties should then be reflected in the metrics applied (see below, parameter metric(s)), and the priorities should be reflected in the ranking when combining multiple metrics that assess different properties.

- Example 1: Find highly accurate liver segmentation algorithm for CT images.
- Example 2: Find lung tumor detection algorithm with high sensitivity and specificity for mammography images.

Corresponding metrics are listed below (parameter metric(s)).

Algorithms should maintain the visual fidelity and structural integrity of reconstructed magnitude images, supporting clinical interpretation. Key metrics include normalized root mean square error (nRMSE), which quantifies intensity differences, and structural similarity index measure (SSIM) and learned perceptual image patch similarity (LPIPS), which evaluates structural and detail-level consistency.

DATA SETS

Data source(s)

a) Specify the device(s) used to acquire the challenge data. This includes details on the device(s) used to acquire the imaging data (e.g. manufacturer) as well as information on additional devices used for performance assessment (e.g. tracking system used in a surgical setting).

The challenge data were acquired on five MRI scanners covering field strengths from 0.1T to 7T. These systems cover ultra-low-field, clinical-field, and ultra-high-field imaging environments.
0.1T: piMR-820H (Point Imaging, China)
1.5T: uMR 670 (United Imaging Healthcare, China)
3T: MAGNETOM Prisma (Siemens Healthineers, Germany)
5T: uMR Jupiter (United Imaging Healthcare, China)
7T: MAGNETOM Terra (Siemens Healthineers, Germany)
Raw imaging data include raw and BIDS compatible files, in alignment with the documentation provided in the ethical protocol.

b) Describe relevant details on the imaging process/data acquisition for each acquisition device (e.g. image acquisition protocol(s)).

Imaging protocols include two major brain MRI contrasts: 3D T1-weighted and 2D T2-weighted FLAIR, acquired across multiple field strengths. All data were processed using a unified pipeline consisting of spatial registration, expert-based quality control, and conservative intensity standardization, in accordance with the approved ethical protocol.

Spatial registration. A cascaded registration strategy implemented in ANTs was used to ensure consistent anatomical alignment across subjects, contrasts, and field strengths. All non-3T images were first aligned to a common 3T T1-weighted reference (Siemens MAGNETOM Prisma) using 3D affine registration. The 3T reference image was subsequently registered to the MNI152 T1 1 mm skull-on template using nonlinear SyN registration, and the resulting transforms were composed and applied in a single resampling step to obtain images directly in MNI space.

For multi-contrast data, nonlinear deformation fields were estimated only from T1-weighted images. Corresponding T2-weighted FLAIR images were first aligned to the subject's T1 image using rigid or affine registration, after which the T1-derived deformation fields were applied, ensuring anatomically consistent normalization across contrasts while avoiding modality-driven overfitting. Linear interpolation was used during resampling.

All images underwent expert quality control to exclude scans with severe artifacts or incomplete coverage. Intensity standardization was applied conservatively to reduce scanner-dependent scaling differences while preserving field-strength-specific contrast characteristics. All datasets are organized according to the BIDS specification with consistent metadata across centers.

c) Specify the center(s)/institute(s) in which the data was acquired and/or the data providing platform/source (e.g. previous challenge). If this information is not provided (e.g. for anonymization reasons), specify why.

The following centers have already contacted us and can support our data collection. Additionally, we are actively reaching out to more international medical centers in the hope of collecting raw data from more sources.

Fudan University Brain Imaging Center, Fudan University, China

Point Imaging, Shanghai, China

Sheshan Campus, Zhongshan Hospital, Fudan University, China

These centers participate in retrospective and prospective MRI data collection across multiple field strengths.

d) Describe relevant characteristics (e.g. level of expertise) of the subjects (e.g. surgeon)/objects (e.g. robot) involved in the data acquisition process (if any).

Data are acquired with a team of radiographers and clinical advisors as appropriate.

Training and test case characteristics

a) State what is meant by one case in this challenge. A case encompasses all data that is processed to produce one result that is compared to the corresponding reference result (i.e. the desired algorithm output).

Examples:

- Training and test cases both represent a CT image of a human brain. Training cases have a weak annotation (tumor present or not and tumor volume (if any)) while the test cases are annotated with the tumor contour (if any).
- A case refers to all information that is available for one particular patient in a specific study. This information always includes the image information as specified in data source(s) (see above) and may include context information (see above). Both training and test cases are annotated with survival (binary) 5 years after (first) image was taken.

In this challenge, one case is defined as a subject–scanner (field strength) unit, i.e., the complete set of imaging data acquired from one subject on one MRI system (one field strength). Each case contains two structural brain MRI contrasts (3D T1w and 2D T2-FLAIR) acquired in the same session under that system's acquisition settings, together with the associated metadata required for processing (e.g., field strength/scanner identifier and sequence information).

For the unpaired multi-field cohort, each case corresponds to a unique subject scanned on a single field strength (100 subjects per scanner; 500 cases in total). For the travelling-volunteer cohort, the same subject is scanned on multiple field strengths; each subject–field acquisition is still treated as an independent case, while the shared subject identifier enables optional within-subject cross-field pairing for evaluation of field-to-field transformation consistency and structural fidelity.

b) State the total number of training, validation and test cases.

The dataset contains 700 cases in total (one case = one subject scanned on one MRI system), comprising 500 unpaired cases and 200 travelling-volunteer cases.
The official split is:
Training set: 500 cases (unpaired cohort only; 100 unique subjects per scanner across five field strengths)
Validation set: 100 cases (travelling-volunteer cohort; 20 subjects × 5 field strengths)
Test set: 100 cases (travelling-volunteer cohort; 20 held-out subjects × 5 field strengths)

The travelling-volunteer cohort is split at the subject level to prevent identity leakage between validation and test. No travelling-volunteer data are included in training, ensuring that model development relies on unpaired multi-field data while evaluation is performed under within-subject paired cross-field comparisons.

c) How much of the data are already annotated (stratified by train test in percentage)?

All retrospective data have already been fully annotated, including both training and test sets, while the prospective data are still being collected.

d) Explain why a total number of cases and the specific proportion of training, validation and test cases was chosen.

The total of 700 cases was chosen to jointly capture (i) realistic between-subject and between-scanner variability across field strengths and (ii) rigorous within-subject paired comparisons for cross-field transformation evaluation. Specifically, the 500-case unpaired cohort (100 unique subjects per scanner across five field strengths) provides sufficient diversity in anatomy, acquisition conditions, and field-dependent image characteristics to support robust model training without relying on paired supervision.

The travelling-volunteer cohort was reserved exclusively for evaluation and split evenly into validation (100 cases; 20 subjects × 5 fields) and test (100 cases; 20 subjects × 5 fields). This design enables statistically stable model selection and final ranking using within-subject cross-field pairing, while enforcing subject-level holdout to prevent identity leakage. By keeping all paired travelling data out of training, the split preserves the intended challenge focus on learning cross-field mappings from unpaired multi-field data, while still providing a controlled and fair benchmark for assessing transformation consistency and structural fidelity across field strengths.

e) Mention further important characteristics of the training, validation and test cases (e.g. class distribution in classification tasks chosen according to real-world distribution vs. equal class distribution) and justify the choice.

The unpaired training cohort intentionally reflects a broad real-world spectrum, including subjects with a wide age range and both healthy volunteers and clinical patients, so that models are exposed to realistic anatomical variability and acquisition heterogeneity. In contrast, the travelling-volunteer cohort used for validation and testing consists of young healthy volunteers, because acquiring within-subject paired scans across five field strengths is logistically demanding, costly, and ethically challenging in patient populations. We therefore use the travelling cohort primarily to provide a controlled paired benchmark for assessing cross-field transformation consistency and structural fidelity, while the diverse training cohort supports robustness to population variability encountered in practice.

f) Challenge organizers are encouraged to (partly) use unseen, unpublished data for their challenges. Describe if new data will be used for the challenge and state the number of cases along with the proportion of new data.

All data used in this challenge are newly collected and have not been previously published. The entire dataset—including both the retrospective cohort (approximately 500 cases) and the prospectively acquired cross-field paired scans (approximately 200 cases)—has been collected and curated exclusively by the organizing team. As a result, 100% of the cases in this challenge consist of unseen, unpublished data, ensuring a fair evaluation setting and preventing any prior exposure or data leakage.

Annotation characteristics

a) Describe the method for determining the reference annotation, i.e. the desired algorithm output. Provide the information separately for the training, validation and test cases if necessary. Possible methods include manual image annotation, in silico ground truth generation and annotation by automatic methods.

If human annotation was involved, state the number of annotators.

This challenge does not involve manual annotation, as the task focuses on image generation and field-to-field synthesis rather than structural delineation or semantic labeling. The reference outputs are therefore defined entirely by the imaging data themselves.

Training cases (unpaired cohort):
No paired reference outputs are provided. Training data consist of real acquired whole-brain MRI across field strengths (including 7T as the target-domain distribution), enabling unpaired/unsupervised learning for 7T-equivalent synthesis.

Validation and test cases (travelling-volunteer cohort; paired):
The reference output is obtained by direct acquisition: for each travelling subject and each contrast, the true 7T

image acquired for the same subject serves as the reference target for Task 1. This provides a within-subject paired benchmark for quantitative comparison of the synthesized 7T-equivalent outputs.

b) Provide the instructions given to the annotators (if any) prior to the annotation. This may include description of a training phase with the software. Provide the information separately for the training, validation and test cases if necessary. Preferably, provide a link to the annotation protocol.

No instructions are required, as this challenge does not involve human annotators or manual labeling. All reference outputs are obtained from standardized MRI acquisition and preprocessing pipelines. Consistency across cases is ensured through unified acquisition guidelines and harmonized preprocessing following BIDS conventions rather than through manual annotation practices.

c) Provide details on the subject(s)/algorithm(s) that annotated the cases (e.g. information on level of expertise such as number of years of professional experience, medically-trained or not). Provide the information separately for the training, validation and test cases if necessary.

This challenge does not include any human annotators. No segmentation, contouring, or manual interpretation is used to generate reference outputs for training, validation, or test datasets. All reference images are derived directly from the corresponding MRI acquisitions.

d) Describe the method(s) used to merge multiple annotations for one case (if any). Provide the information separately for the training, validation and test cases if necessary.

Not applicable. Since no manual annotations are generated and no annotators are involved, there are no multiple annotation sources to merge. All reference outputs are obtained from single-source imaging data collected under standardized acquisition protocols.

Data pre-processing method(s)

Describe the method(s) used for pre-processing the raw training data before it is provided to the participating teams. Provide the information separately for the training, validation and test cases if necessary.

N/A

Sources of error

a) Describe the most relevant possible error sources related to the image annotation. If possible, estimate the magnitude (range) of these errors, using inter-and intra-annotator variability, for example. Provide the information separately for the training, validation and test cases, if necessary.

N.A.

b) In an analogous manner, describe and quantify other relevant sources of error.

N.A.

ASSESSMENT METHODS

Metric(s)

a) Define the metric(s) to assess a property of an algorithm. These metrics should reflect the desired algorithm properties described in assessment aim(s) (see above). State which metric(s) were used to compute the ranking(s) (if any).

- Example 1: Dice Similarity Coefficient (DSC)
- Example 2: Area under curve (AUC)

The assessment methodology for Task 2 focuses on evaluating image fidelity, anatomical consistency, and robustness to image degradations. To assess the quality of generated higher-field images from ultra-low-field (0.1T) inputs, the following complementary metrics are employed:

Normalized Root Mean Square Error (nRMSE): This metric quantifies global intensity and contrast deviations between the generated images and higher-field reference images, providing a measure of numerical reconstruction accuracy under severe noise and resolution limitations.

Structural Similarity Index Measure (SSIM): SSIM evaluates structural and perceptual similarity between generated and reference images, with particular sensitivity to the preservation of anatomically meaningful boundaries that are often degraded at ultra-low field strengths.

Learned Perceptual Image Patch Similarity (LPIPS): LPIPS measures perceptual similarity based on deep feature representations and is sensitive to high-level structural and textural discrepancies, enabling assessment of visually and anatomically plausible recovery under extreme degradation.

Dice overlap (deep gray matter): Dice similarity is computed on automated SynthSeg-derived segmentations of 14 bilateral deep gray matter nuclei. Dice quantifies spatial overlap and boundary consistency between synthesized images and paired higher-field references, providing anatomically grounded assessment of regional structural preservation under extreme ultra-low-field degradation.

Normalized volume consistency (deep gray matter): For each deep gray matter structure, volume consistency is used. This bounded consistency score captures relative volumetric deviation and complements Dice by detecting systematic expansion or contraction not reflected by spatial overlap alone.

Together, these metrics provide complementary perspectives on translation accuracy, structural fidelity, and perceptual realism in the challenging ultra-low-field MRI regime. Metrics used for ranking will be specified in the evaluation guideline provided to participants.

b) Justify why the metric(s) was/were chosen, preferably with reference to the biomedical application.

The chosen metrics reflect the essential biomedical requirements of this challenge. nRMSE quantifies global intensity and contrast agreement, ensuring that synthesized images accurately reproduce field-dependent signal characteristics. SSIM evaluates perceptual and anatomical fidelity, capturing the preservation of cortical boundaries, deep gray nuclei, and other fine structural details. LPIPS complements these measures by assessing high-level perceptual similarity that aligns with human visual judgement, providing sensitivity to subtle textural and contrast variations important for neuroanatomical interpretation. Dice overlap on deep gray matter nuclei provides anatomically grounded validation of spatial consistency and boundary preservation in clinically meaningful subcortical structures, which are particularly sensitive to synthesis-induced distortions under extreme field transitions. Normalized volume consistency complements Dice by quantifying relative volumetric deviation, capturing systematic expansion or contraction of structures that may not be reflected by spatial overlap alone.Combined, these five metrics offer a balanced evaluation of numerical accuracy, perceptual realism, and anatomically grounded structural preservation for field-aware MRI generation.

Ranking method(s)

a) Describe the method used to compute a performance rank for all submitted algorithms based on the generated metric results on the test cases. Typically the text will describe how results obtained per case and metric are aggregated to arrive at a final score/ranking. Ideally, provide the ranking scheme as a concrete pseudo code.

The final ranking of all submitted algorithms is based on their overall performance across all test cases, as follows:
1. Individual Metric Ranking: For each metric (nRMSE, SSIM, and LPIPS), the metric is first computed for every test case and then averaged across all test cases. Lower values are preferred for nRMSE and LPIPS, while higher values are preferred for SSIM. Each algorithm is ranked independently for each metric based on its mean performance.
2. Overall Ranking: The overall ranking is obtained by summing the per-metric ranks for each algorithm. Algorithms with the lowest composite scores achieve the highest final rank. This rank-summation approach ensures that differences in metric scales do not influence the overall evaluation. This procedure yields a stable and interpretable ranking that reflects performance across complementary dimensions of image quality.

b) Describe the method(s) used to manage submissions with missing results on test cases.

Participating teams are required to submit Docker containers and process all the cases in the test set on our server. For the cases without valid output, we will assign it to the lowest value of each metric.

c) Justify why the described ranking scheme(s) was/were used.

The composite rank-summation scheme is chosen to ensure fair and balanced evaluation across multiple complementary aspects of image quality. nRMSE measures numerical agreement, SSIM captures structural fidelity, LPIPS reflects perceptual realism, while Dice overlap and normalized volume consistency explicitly assess regional deep gray matter preservation. By ranking algorithms separately for each metric and combining ranks rather than raw metric values, the method avoids biases arising from differences in metric scale or dynamic range. This multi-metric ranking approach provides a robust and clinically meaningful assessment of algorithms designed for field-aware MRI generation.

Statistical analyses

Provide an overview of the statistical approaches used in the scope of the challenge analysis. Details can be provided in the parameters below. For each parameter, justify why the described statistical method(s) was/were used and, if necessary, add a description of any method used to assess whether the data met the assumptions required for the particular statistical approach.

The statistical analysis framework of MRIxFields2026 is designed to ensure robust, unbiased, and clinically meaningful evaluation of cross-field MRI generation algorithms. Given the heterogeneous nature of the dataset—spanning magnetic field strengths from 0.1T to 7T, multiple scanners, and both paired and unpaired acquisitions—the challenge adopts case-wise, distribution-aware, and rank-robust statistical methods. Performance is primarily assessed on held-out, unseen test cases, with all metrics computed at the individual-case level and subsequently aggregated to support population-level inference. Where applicable, non-parametric statistics are preferred to avoid assumptions of normality, and all analyses are conducted under a unified evaluation protocol to ensure comparability across tasks and teams.

Provide a description of how the precision of the performance estimates of individual algorithms is assessed (e.g. confidence interval of the mean on the test set computed using percentile bootstrap, confidence interval of the accuracy on the test set computed using percentile bootstrap).

The precision of performance estimates for each submitted algorithm is quantified using bootstrap-based confidence intervals. Specifically, for each evaluation metric, percentile bootstrap resampling (e.g., 1,000 resamples with replacement at the case level) is applied to the test set to estimate 95% confidence intervals of the mean performance. This approach is well suited to the challenge setting, as it does not rely on distributional assumptions and is robust to skewed or heavy-tailed metric distributions that commonly arise in cross-domain MRI evaluation. Confidence intervals are reported alongside point estimates to reflect uncertainty and to enable fair comparison between methods.

Provide a description of how variability of the performance of individual algorithms across tests cases is assessed (e.g. SD across test cases, IQR, graphs, reporting outliers...).

To characterize how algorithm performance varies across individual test cases, case-wise variability analyses are performed. For each method and metric, variability is summarized using the standard deviation (SD) and interquartile range (IQR) across test cases, providing complementary views of dispersion and robustness. In addition, case-level performance distributions (e.g., boxplots or violin plots) are used to visualize heterogeneity, identify outliers, and highlight failure modes under specific field strengths or acquisition conditions.

Provide a description of how variability of rankings is assessed.

To evaluate the stability of algorithm rankings, variability across test cases is quantified using standard measures such as the standard deviation and interquartile range of per-metric performance. Pairwise rank correlations are computed to assess agreement between rankings derived from nRMSE, SSIM, and LPIPS. These analyses ensure that the final composite ranking is robust to minor fluctuations in individual metric values.

Provide a description of statistical tests that are used to assess whether the differences in performance between algorithms are statistically significant.

To assess whether observed performance differences between algorithms are statistically significant, paired, non-parametric hypothesis tests are employed. For pairwise comparisons between methods on the same test cases, the Wilcoxon signed-rank test is used, as it accounts for within-case pairing and does not assume normality. When comparing more than two algorithms simultaneously, a Friedman test is applied, followed by appropriate post-hoc pairwise comparisons with multiple-comparison correction (e.g., Holm–Bonferroni adjustment).

Provide a description of the missing data handling.

All test cases are verified for completeness prior to release, and missing data are not expected. If a submission fails to produce an output for a specific case, that case will be assigned the lowest possible metric score as defined in the ranking procedure, ensuring consistent treatment across teams.

Indicate any software product that is used for all data analysis methods.

All statistical analyses and visualizations are performed using Python-based tools, including NumPy, SciPy, and Matplotlib. These libraries support rank correlation analysis, variability assessment, and aggregate metric evaluation needed for the challenge.

Further analyses

Present further analyses to be performed (if applicable), e.g. related to

- combining algorithms via ensembling,
- inter-algorithm variability,
- common problems/biases of the submitted methods, or
- ranking variability.

N/A

TASK 3: Controllable Field-to-Field MRI Synthesis with a Unified Conditional Model

SUMMARY

Abstract

Provide a summary of the challenge purpose. This should include a general introduction in the topic from both a biomedical as well as from a technical point of view and clearly state the envisioned technical and/or biomedical impact of the challenge.

Magnetic resonance imaging systems operate across a wide spectrum of magnetic field strengths, each characterized by distinct signal behavior, contrast mechanisms, spatial resolution, and acquisition constraints. While fixed-target high-field synthesis and ultra-low-field enhancement address important and complementary use cases, many practical scenarios require flexible and controllable transformation between arbitrary MRI field strengths, such as harmonizing heterogeneous datasets or adapting image generation to changing scanner availability. In contemporary multi-center studies and real-world clinical workflows, MRI datasets are often collected across multiple field strengths without a single canonical reference domain, and acquisition configurations may evolve over time due to hardware upgrades, protocol changes, or site-specific constraints. Under such conditions, learning separate pairwise mappings or relying on fixed-direction translation pipelines becomes increasingly inefficient and difficult to maintain. Moreover, the lack of explicit controllability limits the ability of generative models to adapt outputs to user-specified field targets or to generalize to previously unseen field combinations.

This task focuses on controllable field-to-field MRI synthesis using a single unified conditional model, where both the input and target field strengths can be explicitly specified. Evaluation follows an any-to-any protocol with stratified field-pair analysis, explicitly assessing performance across diverse and extreme transitions (e.g., 0.1T7T). Unlike pairwise or fixed-direction translation, the objective is to learn a generalizable and field-aware representation that enables dynamic image generation conditioned on desired target field parameters. Models are expected to capture field-dependent variations in contrast behavior, noise characteristics, and spatial resolution, while preserving anatomical fidelity across diverse transformation settings.

The availability of multi-field data supports systematic evaluation of synthesis accuracy and controllability across a wide range of field-to-field mappings, including both paired and unpaired acquisition scenarios. Assessment combines voxel-level similarity metrics (nRMSE, SSIM, LPIPS) together with anatomically grounded segmentation measures, including mean Dice overlap and normalized volume consistency across deep gray matter nuclei, enabling quantitative evaluation of both perceptual realism and regional structural preservation.By explicitly requiring conditional control within a unified modeling framework, this task evaluates whether generative MRI models can scale beyond isolated translation tasks and support generalizable, controllable, and extensible field-aware image synthesis. Together with fixed-target high-field generation and ultra-low-field enhancement, Task 3 completes the challenge by benchmarking the capability of generative models to integrate flexibility, controllability, and robustness within a single coherent framework suitable for scalable multi-center harmonization and adaptive clinical deployment.

Keywords

List the primary keywords that characterize the task.

Controllable MRI synthesis, Conditioned generation, Harmonization, Generative modeling

ORGANIZATION

Organizers

a) Provide information on the organizing team (names and affiliations).

Chengyan Wang
Human Phenome Institute, Fudan University, China

Fangrong Zong, Yong Liu
School of Artificial Intelligence, Beijing University of Posts and Telecommunications, China

Weirui Cai, Xin Guo, Chang Xu
Institute of Science and Technology for Brain-inspired Intelligence, Fudan University, China

Jun Lyu
School of Computer and Control Engineering, Yantai University, China

Andrew Web
Leiden University, Netherlands

Tianxing He
Department of Biomedical Engineering, Fudan University, China

Yuntong Lyu
School of Medicine, Fudan University, China

b) Provide information on the primary contact person.

Fangrong Zong, Fangrong.zong@bupt.edu.cn
School of Artificial Intelligence, Beijing University of Posts and Telecommunications, China

c) Indicate whether clinicians are part of the organizing team. If yes, describe their role.

Yes.
The organizing team includes clinicians with expertise in neuroradiology, clinical neuroscience, and brain MRI, who play a central role in ensuring the clinical relevance, anatomical validity, and translational value of the challenge. They contribute to the design of acquisition protocols across different field strengths, and oversee quality assurance of retrospective and prospectively acquired scans. Clinicians guide the definition of task objectives from a clinical perspective, particularly in relation to high-field-equivalent image fidelity, ultra-low-field enhancement for diagnostic usability, and cross-field harmonization in multi-center settings.Through their involvement, the clinical organizers help anchor the benchmark in realistic workflows and ensure that the developed methods can support multi-center neuroscience research and future clinical deployment across heterogeneous MRI systems.

Life cycle type

Define the intended submission cycle of the challenge. Include information on whether/how the challenge will be continued after the challenge has taken place.Not every challenge closes after the submission deadline (one-time event). Sometimes it is possible to submit results after the deadline (open call) or the challenge is repeated with some modifications (repeated event).

Examples:

- One-time event with fixed conference submission deadline
- Open call (challenge opens for new submissions after conference deadline)
- Repeated event with annual fixed conference submission deadline

One-time event with fixed conference submission deadline

Challenge venue and platform

a) Report the event (e.g. conference) that is associated with the challenge (if any).

29th International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI 2026).

b) Report the platform used to run the challenge.

synapse.org

c) Do you agree that the your submission is shared with the platform (e.g., grand-challenge, synapse...) that you indicated?

Please note: 1) this purpose of such sharing is that the challenge chairs and the platform can communicate smoothly, your answer won't impact the review of your proposal; 2) regardless of your response to this question, it is your responsibility to perform all actions required by the platform (e.g. filling their submission request).

Yes

d) Provide the URL for the challenge website (if any).

Website for MRIxFields2026: https://mrixfields.github.io/2026; Synapse platform for MRIxFields2026: https://www.synapse.org/Synapse:syn72060672

Participation policies

a) Define the allowed user interaction of the algorithms assessed. This includes the policy regarding any curation, (pre-)processing and (pre-)training steps.

Fully automatic

b) Define the policy on the usage of training data. The data used to train algorithms may, for example, be restricted to the data provided by the challenge or may also include publicly available data including (open) pre-trained nets. Clarify whether such additional data needs to be publicly available at the time of the challenge launch. Clarify whether adding (private) annotations of the public data is allowed.

It should be restricted to the data provided by the current MRIxFields2026 challenges, under the terms and conditions associated with the data usage.

c) Define the participation policy for members of the organizers' institutes. For example, members of the organizers' institutes may participate in the challenge but are not eligible for awards.

May participate but not eligible for awards and not listed in leaderboard

d) Define the award policy. In particular, provide details with respect to challenge prizes.

The top 5 winners receive monetary awards. We are in active negotiation with the sponsor about the value (approximately $2000 in total).

e) Define the policy for result announcement.

Examples:

- Top 3 performing methods will be announced publicly.
- Participating teams can choose whether the performance results will be made public.

All submissions will be reported in the leaderboard. Participating teams can opt out of publication of their results in the leaderboard.

Prize-winning methods will be announced publicly as part of a scientific session at the MICCAI annual meeting.

f) Define the publication policy. In particular, provide details on ...

- ... who of the participating teams/the participating teams' members qualifies as author
- ... whether the participating teams may publish their own results separately, and (if so)
- ... whether an embargo time is defined (so that challenge organizers can publish a challenge paper first).

Participating teams with a valid submission can nominate their team members as co-authors for the challenge paper. We reserve the right to exclude teams if they break the challenge rules. Participating teams can publish their own results but after a 3-month embargo period.

Submission method

a) Describe the method used for result submission. Preferably, provide a link to the submission instructions.

Examples:

- Docker container on the Synapse platform. Link to submission instructions: <URL>
- Algorithm output was sent to organizers via e-mail. Submission instructions were sent by e-mail.

Docker container will be accepted as submission through the Synapse platform. Submission details will be published at the time point of challenge announcement.

b) Provide information on the possibility for participating teams to evaluate their algorithms before submitting final results. For example, many challenges allow submission of multiple results, and only the last run is officially counted to compute challenge results.

Participating teams are allowed to make 3 formal submissions per task. Only the last run submission is officially counted to rank challenge results. Before the final submission on the test set, participants can test their Docker containers on the validation dataset to avoid submission errors.

Challenge schedule

Provide a timetable for the challenge. Preferably, this should include

- the release date(s) of the training cases (if any)
- the registration date/period
- the release date(s) of the test cases and validation cases (if any)
- the submission date(s)
- associated workshop days (if any)
- the release date(s) of the results

[Apr 01, 2026] website opens for registration, release training and validation images
[May 20, 2026] submission system opens for validation
[Aug 01, 2026] submission system opens for testing
[Sept 10, 2026] registration and docker submission deadline
[Oct 08, 2026] release final results during the MICCAI annual meeting

Ethics approval

Indicate whether ethics approval is necessary for the data. If yes, provide details on the ethics approval, preferably institutional review board, location, date and number of the ethics approval (if applicable). Add the URL or a reference to the document of the ethics approval (if available).

We are currently applying for ethical approval, which is expected to be granted within approximately one month.

Data usage agreement

Clarify how the data can be used and distributed by the teams that participate in the challenge and by others during and after the challenge. This should include the explicit listing of the license applied.

Examples:

- CC BY (Attribution)
- CC BY-SA (Attribution-ShareAlike)
- CC BY-ND (Attribution-NoDerivs)
- CC BY-NC (Attribution-NonCommercial)
- CC BY-NC-SA (Attribution-NonCommercial-ShareAlike)
- CC BY-NC-ND (Attribution-NonCommercial-NoDerivs)

Please note that the data license should not differ among sources. In case a license has to be changed, it has to be reported to the MICCAI challenges team and changed in the proposal.

CC BY-NC (Attribution-NonCommercial)

Code availability

a) Provide information on the accessibility of the organizers' evaluation software (e.g. code to produce rankings). Preferably, provide a link to the code and add information on the supported platforms.

We will provide the synthesis evaluation code to participants before the challenge officially begins. This code will enable participants to test their algorithms and understand the evaluation process.

b) In an analogous manner, provide information on the accessibility of the participating teams' code.

The participating teams are suggested to provide links to their code on our website for reproducibility study. However, it is not a condition of participation.

Conflicts of interest

Provide information related to conflicts of interest. In particular provide information related to sponsoring/funding of the challenge. Also, state explicitly who had/will have access to the test case labels and when.

We declare no conflicts of interest. Test images will only be accessible to the challenge organizers.

MISSION OF THE CHALLENGE

Field(s) of application

State the main field(s) of application that the participating algorithms target.

Examples:

- Diagnosis
- Education
- Intervention assistance
- Intervention follow-up
- Intervention planning
- Prognosis
- Research
- Screening
- Training
- Cross-phase

Research,Diagnosis

Task category(ies)

State the task category(ies)

Examples:

- Classification
- Detection
- Localization
- Modeling

- Prediction
- Reconstruction
- Registration
- Retrieval
- Segmentation
- Tracking

Reconstruction

Cohorts

We distinguish between the target cohort and the challenge cohort. For example, a challenge could be designed around the task of medical instrument tracking in robotic kidney surgery. While the challenge could be based on ex vivo data obtained from a laparoscopic training environment with porcine organs (challenge cohort), the final biomedical application (i.e. robotic kidney surgery) would be targeted on real patients with certain characteristics defined by inclusion criteria such as restrictions regarding sex or age (target cohort).

a) Describe the target cohort, i.e. the subjects/objects from whom/which the data would be acquired in the final biomedical application.

The target cohort for this challenge consists of individuals undergoing brain MRI examinations across a broad range of clinical and research environments. These include patients who may benefit from improved image quality in settings where high-field MRI is unavailable, as well as individuals undergoing neurological evaluation for conditions in which enhanced structural visibility, contrast fidelity, or multi-field harmonization could support diagnosis or longitudinal assessment. The envisioned applications span accessible imaging in low-resource settings, advanced neuroimaging in ultra-high-field environments, and multi-center studies requiring consistent image quality despite variability in scanner hardware and field strength. The ultimate goal is to enable reliable field-aware image generation and enhancement methods that improve the fidelity and comparability of brain MRI for diverse patient populations.

b) Describe the challenge cohort, i.e. the subject(s)/object(s) from whom/which the challenge data was acquired.

The challenge cohort consists of healthy volunteers who underwent brain MRI at five field strengths (0.1T, 1.5T, 3T, 5T, and 7T). The dataset is organized into two complementary components:
1.Unpaired multi-field imaging cohort (non-overlapping subjects):
A total of 500 independent subject–scanner cases, collected as 100 unique healthy subjects per scanner across the five MRI systems. Each case includes two standard brain MRI contrasts acquired under the corresponding system settings. This cohort captures realistic inter-scanner and field-dependent variability in signal-to-noise ratio, contrast behavior, and spatial resolution, while avoiding subject-level pairing.

2.Travelling-volunteer imaging cohort (paired cross-field scans):
A dedicated cohort of 40 travelling volunteers were scanned on all five MRI systems, yielding 200 paired subject–scanner cases (40 subjects × 5 field strengths). This component provides within-subject cross-field pairing for benchmarking field-to-field transformation consistency and structural fidelity.

No clinical disease labels are provided. Together, the unpaired and paired cohorts enable evaluation of generative models for high-field synthesis, ultra-low-field enhancement, and controllable field-to-field transformation, under both realistic population variability and controlled within-subject comparisons.

Imaging modality(ies)

Specify the imaging technique(s) applied in the challenge.

Magnetic Resonance Imaging

Context information

Provide additional information given along with the images. The information may correspond ...

a) ... directly to the image data (e.g. tumor volume).

None

b) ... to the patient in general (e.g. sex, medical history).

None

Target entity(ies)

a) Describe the data origin, i.e. the region(s)/part(s) of subject(s)/object(s) from whom/which the image data would be acquired in the final biomedical application (e.g. brain shown in computed tomography (CT) data, abdomen shown in laparoscopic video data, operating room shown in video data, thorax shown in fluoroscopy video). If necessary, differentiate between target and challenge cohort.

The data for this challenge originates from brain MRI acquired across a wide range of magnetic field strengths, covering the entire brain including cortical and subcortical regions, deep gray matter structures, and white matter. The imaging focuses exclusively on two commonly used structural MRI contrasts, namely T1-weighted and T2-weighted FLAIR, which are widely adopted in both clinical practice and neuroscience research and exhibit distinct contrast behavior across field strengths.

The challenge cohort consists of healthy volunteers scanned at multiple field strengths between 0.1T and 7T and is composed of two complementary data subsets. The first subset comprises retrospectively collected multi-field cases, acquired from different individuals at different field strengths and therefore non-paired across fields, reflecting realistic variability in acquisition conditions, noise characteristics, contrast behavior, and spatial resolution. The second subset includes prospectively acquired cross-field paired scans, in which the same subjects were imaged at multiple field strengths under controlled protocols. Together, these retrospective and prospective data enable systematic evaluation of generative models for high-field synthesis, ultra-low-field enhancement, and controllable field-to-field transformation across diverse MRI domains.

b) Describe the algorithm target, i.e. the structure(s)/subject(s)/object(s)/component(s) that the participating algorithms have been designed to focus on (e.g. tumor in the brain, tip of a medical instrument, nurse in an operating theater, catheter in a fluoroscopy scan). If necessary, differentiate between target and challenge cohort.

The algorithms in this challenge are designed to generate or enhance brain MRI across different field strengths while preserving anatomical fidelity and contrast-specific tissue characteristics. Participants' models are expected to recover fine anatomical detail, restore tissue contrast, and produce consistent structural representations that align with the appearance of higher-field imaging. Depending on the specific task, the algorithms may generate 7T-equivalent images from arbitrary field strengths, enhance ultra-low-field acquisitions toward higher-field quality, or perform controllable transformations between any pair of field strengths using a unified conditional model. Across all tasks, the goal is to achieve anatomically accurate and contrast-consistent outputs that improve comparability across field strengths and support reliable downstream neuroimaging analyses.

Assessment aim(s)

Identify the property(ies) of the algorithms to be optimized to perform well in the challenge. If multiple properties are assessed, prioritize them (if appropriate). The properties should then be reflected in the metrics applied (see below, parameter metric(s)), and the priorities should be reflected in the ranking when combining multiple metrics that assess different properties.

- Example 1: Find highly accurate liver segmentation algorithm for CT images.
- Example 2: Find lung tumor detection algorithm with high sensitivity and specificity for mammography images.

Corresponding metrics are listed below (parameter metric(s)).

Algorithms should maintain the visual fidelity and structural integrity of reconstructed magnitude images, supporting clinical interpretation. Key metrics include normalized root mean square error (nRMSE), which quantifies intensity differences, and structural similarity index measure (SSIM), learned perceptual image patch similarity (LPIPS), Dice overlap, and normalized volume consistency, which evaluates structural and detail-level consistency.

DATA SETS

Data source(s)

a) Specify the device(s) used to acquire the challenge data. This includes details on the device(s) used to acquire the imaging data (e.g. manufacturer) as well as information on additional devices used for performance assessment (e.g. tracking system used in a surgical setting).

The challenge data were acquired on five MRI scanners covering field strengths from 0.1T to 7T. These systems cover ultra-low-field, clinical-field, and ultra-high-field imaging environments.
0.1T: piMR-820H (Point Imaging, China)
1.5T: uMR 670 (United Imaging Healthcare, China)
3T: MAGNETOM Prisma (Siemens Healthineers, Germany)
5T: uMR Jupiter (United Imaging Healthcare, China)
7T: MAGNETOM Terra (Siemens Healthineers, Germany)
Raw imaging data include raw and BIDS compatible files, in alignment with the documentation provided in the ethical protocol.

b) Describe relevant details on the imaging process/data acquisition for each acquisition device (e.g. image acquisition protocol(s)).

Imaging protocols include two major brain MRI contrasts: 3D T1-weighted and 2D T2-weighted FLAIR, acquired across multiple field strengths. All data were processed using a unified pipeline consisting of spatial registration, expert-based quality control, and conservative intensity standardization, in accordance with the approved ethical protocol.

Spatial registration. A cascaded registration strategy implemented in ANTs was used to ensure consistent anatomical alignment across subjects, contrasts, and field strengths. All non-3T images were first aligned to a common 3T T1-weighted reference (Siemens MAGNETOM Prisma) using 3D affine registration. The 3T reference image was subsequently registered to the MNI152 T1 1 mm skull-on template using nonlinear SyN registration, and the resulting transforms were composed and applied in a single resampling step to obtain images directly in MNI space.

For multi-contrast data, nonlinear deformation fields were estimated only from T1-weighted images. Corresponding T2-weighted FLAIR images were first aligned to the subject's T1 image using rigid or affine registration, after which the T1-derived deformation fields were applied, ensuring anatomically consistent normalization across contrasts while avoiding modality-driven overfitting. Linear interpolation was used during resampling.

All images underwent expert quality control to exclude scans with severe artifacts or incomplete coverage. Intensity standardization was applied conservatively to reduce scanner-dependent scaling differences while preserving field-strength-specific contrast characteristics. All datasets are organized according to the BIDS specification with consistent metadata across centers.

c) Specify the center(s)/institute(s) in which the data was acquired and/or the data providing platform/source (e.g. previous challenge). If this information is not provided (e.g. for anonymization reasons), specify why.

The following centers have already contacted us and can support our data collection. Additionally, we are actively reaching out to more international medical centers in the hope of collecting raw data from more sources.

Fudan University Brain Imaging Center, Fudan University, China

Point Imaging, Shanghai, China

Sheshan Campus, Zhongshan Hospital, Fudan University, China

These centers participate in retrospective and prospective MRI data collection across multiple field strengths.

d) Describe relevant characteristics (e.g. level of expertise) of the subjects (e.g. surgeon)/objects (e.g. robot) involved in the data acquisition process (if any).

Data are acquired with a team of radiographers and clinical advisors as appropriate.

Training and test case characteristics

a) State what is meant by one case in this challenge. A case encompasses all data that is processed to produce one result that is compared to the corresponding reference result (i.e. the desired algorithm output).

Examples:

- Training and test cases both represent a CT image of a human brain. Training cases have a weak annotation (tumor present or not and tumor volume (if any)) while the test cases are annotated with the tumor contour (if any).
- A case refers to all information that is available for one particular patient in a specific study. This information always includes the image information as specified in data source(s) (see above) and may include context

information (see above). Both training and test cases are annotated with survival (binary) 5 years after (first) image was taken.

In this challenge, one case is defined as a subject–scanner (field strength) unit, i.e., the complete set of imaging data acquired from one subject on one MRI system (one field strength). Each case contains two structural brain MRI contrasts (3D T1w and 2D T2-FLAIR) acquired in the same session under that system's acquisition settings, together with the associated metadata required for processing (e.g., field strength/scanner identifier and sequence information).

For the unpaired multi-field cohort, each case corresponds to a unique subject scanned on a single field strength (100 subjects per scanner; 500 cases in total). For the travelling-volunteer cohort, the same subject is scanned on multiple field strengths; each subject–field acquisition is still treated as an independent case, while the shared subject identifier enables optional within-subject cross-field pairing for evaluation of field-to-field transformation consistency and structural fidelity.

b) State the total number of training, validation and test cases.

The dataset contains 700 cases in total (one case = one subject scanned on one MRI system), comprising 500 unpaired cases and 200 travelling-volunteer cases.
The official split is:
Training set: 500 cases (unpaired cohort only; 100 unique subjects per scanner across five field strengths)
Validation set: 100 cases (travelling-volunteer cohort; 20 subjects × 5 field strengths)
Test set: 100 cases (travelling-volunteer cohort; 20 held-out subjects × 5 field strengths)

The travelling-volunteer cohort is split at the subject level to prevent identity leakage between validation and test. No travelling-volunteer data are included in training, ensuring that model development relies on unpaired multi-field data while evaluation is performed under within-subject paired cross-field comparisons.

c) How much of the data are already annotated (stratified by train test in percentage)?

All retrospective data have already been fully annotated, including both training and test sets, while the prospective data are still being collected.

d) Explain why a total number of cases and the specific proportion of training, validation and test cases was chosen.

The total of 700 cases was chosen to jointly capture (i) realistic between-subject and between-scanner variability across field strengths and (ii) rigorous within-subject paired comparisons for cross-field transformation evaluation. Specifically, the 500-case unpaired cohort (100 unique subjects per scanner across five field strengths) provides sufficient diversity in anatomy, acquisition conditions, and field-dependent image characteristics to support robust model training without relying on paired supervision.

The travelling-volunteer cohort was reserved exclusively for evaluation and split evenly into validation (100 cases; 20 subjects × 5 fields) and test (100 cases; 20 subjects × 5 fields). This design enables statistically stable model selection and final ranking using within-subject cross-field pairing, while enforcing subject-level holdout to prevent identity leakage. By keeping all paired travelling data out of training, the split preserves the intended challenge focus on learning cross-field mappings from unpaired multi-field data, while still providing a controlled and fair benchmark for assessing transformation consistency and structural fidelity across field strengths.

e) Mention further important characteristics of the training, validation and test cases (e.g. class distribution in classification tasks chosen according to real-world distribution vs. equal class distribution) and justify the choice.

The unpaired training cohort intentionally reflects a broad real-world spectrum, including subjects with a wide age range and both healthy volunteers and clinical patients, so that models are exposed to realistic anatomical variability and acquisition heterogeneity. In contrast, the travelling-volunteer cohort used for validation and testing consists of young healthy volunteers, because acquiring within-subject paired scans across five field strengths is logistically demanding, costly, and ethically challenging in patient populations. We therefore use the travelling cohort primarily to provide a controlled paired benchmark for assessing cross-field transformation consistency and structural fidelity, while the diverse training cohort supports robustness to population variability encountered in practice.

f) Challenge organizers are encouraged to (partly) use unseen, unpublished data for their challenges. Describe if new data will be used for the challenge and state the number of cases along with the proportion of new data.

All data used in this challenge are newly collected and have not been previously published. The entire dataset—including both the retrospective cohort (approximately 500 cases) and the prospectively acquired cross-field paired scans (approximately 200 cases)—has been collected and curated exclusively by the organizing team. As a result, 100% of the cases in this challenge consist of unseen, unpublished data, ensuring a fair evaluation setting and preventing any prior exposure or data leakage.

Annotation characteristics

a) Describe the method for determining the reference annotation, i.e. the desired algorithm output. Provide the information separately for the training, validation and test cases if necessary. Possible methods include manual image annotation, in silico ground truth generation and annotation by automatic methods.

If human annotation was involved, state the number of annotators.

This challenge does not involve manual annotation, as the task focuses on image generation and field-to-field synthesis rather than structural delineation or semantic labeling. The reference outputs are therefore defined entirely by the imaging data themselves.

Training cases (unpaired cohort):
No paired reference outputs are provided. Training data consist of real acquired whole-brain MRI across field strengths (including 7T as the target-domain distribution), enabling unpaired/unsupervised learning for 7T-equivalent synthesis.

Validation and test cases (travelling-volunteer cohort; paired):
The reference output is obtained by direct acquisition: for each travelling subject and each contrast, the true 7T image acquired for the same subject serves as the reference target for Task 1. This provides a within-subject paired benchmark for quantitative comparison of the synthesized 7T-equivalent outputs.

b) Provide the instructions given to the annotators (if any) prior to the annotation. This may include description of a training phase with the software. Provide the information separately for the training, validation and test cases if necessary. Preferably, provide a link to the annotation protocol.

No instructions are required, as this challenge does not involve human annotators or manual labeling. All reference outputs are obtained from standardized MRI acquisition and preprocessing pipelines. Consistency across cases is ensured through unified acquisition guidelines and harmonized preprocessing following BIDS conventions rather than through manual annotation practices.

c) Provide details on the subject(s)/algorithm(s) that annotated the cases (e.g. information on level of expertise such as number of years of professional experience, medically-trained or not). Provide the information separately for the training, validation and test cases if necessary.

This challenge does not include any human annotators. No segmentation, contouring, or manual interpretation is used to generate reference outputs for training, validation, or test datasets. All reference images are derived directly from the corresponding MRI acquisitions.

d) Describe the method(s) used to merge multiple annotations for one case (if any). Provide the information separately for the training, validation and test cases if necessary.

Not applicable. Since no manual annotations are generated and no annotators are involved, there are no multiple annotation sources to merge. All reference outputs are obtained from single-source imaging data collected under standardized acquisition protocols.

Data pre-processing method(s)

Describe the method(s) used for pre-processing the raw training data before it is provided to the participating teams. Provide the information separately for the training, validation and test cases if necessary.

N/A

Sources of error

a) Describe the most relevant possible error sources related to the image annotation. If possible, estimate the magnitude (range) of these errors, using inter-and intra-annotator variability, for example. Provide the information separately for the training, validation and test cases, if necessary.

N.A.

b) In an analogous manner, describe and quantify other relevant sources of error.

N.A.

ASSESSMENT METHODS

Metric(s)

a) Define the metric(s) to assess a property of an algorithm. These metrics should reflect the desired algorithm properties described in assessment aim(s) (see above). State which metric(s) were used to compute the ranking(s) (if any).

- Example 1: Dice Similarity Coefficient (DSC)
- Example 2: Area under curve (AUC)

The assessment methodology for Task 3 is designed to evaluate image fidelity, field-awareness, and controllability in field-conditioned MRI synthesis. To assess the quality and correctness of generated images under arbitrary field-to-field transformations, the following metrics are employed:

Normalized Root Mean Square Error (nRMSE): nRMSE measures global intensity and contrast differences between synthesized images and reference images at the specified target field strength, serving as a measure of numerical accuracy for field-conditioned generation.

Structural Similarity Index Measure (SSIM): SSIM evaluates structural and perceptual similarity between generated images and target-field references, capturing the preservation of anatomical structures under varying field-conditioning inputs.

Learned Perceptual Image Patch Similarity (LPIPS): LPIPS assesses perceptual similarity based on deep feature representations and is sensitive to high-level textural and structural differences that reflect field-dependent image characteristics.

Dice overlap (deep gray matter): Dice similarity is computed on automated SynthSeg-derived segmentations of 14 bilateral deep gray matter nuclei . Dice quantifies spatial overlap and boundary consistency between synthesized images and target-field references, providing anatomically grounded assessment of regional structural preservation under arbitrary field-to-field transformations.

Normalized volume consistency (deep gray matter): For each deep gray matter structure, volume consistency is used. This bounded consistency score captures relative volumetric deviation and complements Dice by detecting systematic expansion or contraction not reflected by spatial overlap alone.

To explicitly assess controllability, these metrics are computed conditioned on different target field specifications, enabling comparison of outputs generated from the same input image under varying field-conditioning parameters. These five complementary metrics (nRMSE, SSIM, LPIPS, mean Dice overlap, and mean normalized volume consistency) jointly define the primary leaderboard via rank-summation, as specified in the evaluation guideline provided to participants.

b) Justify why the metric(s) was/were chosen, preferably with reference to the biomedical application.

The selected metrics reflect the biomedical and methodological objectives of Task 3, which emphasizes controllable and field-aware MRI generation. nRMSE quantifies whether synthesized images accurately reproduce the global signal and contrast properties associated with the specified target field strength. SSIM assesses the preservation of anatomical structures while adapting field-dependent imaging characteristics, ensuring that controllability does not compromise structural integrity. LPIPS complements these measures by capturing perceptual differences that align with human judgement of field-specific image appearance. Dice overlap on deep gray matter nuclei provides anatomically grounded validation of spatial consistency in clinically meaningful subcortical structures, which are particularly sensitive to synthesis-induced distortions under field transitions. Normalized volume consistency complements Dice by quantifying relative volumetric bias, capturing systematic expansion or contraction that may not be reflected by spatial overlap alone.

By evaluating these metrics across multiple target field conditions for identical input images, the assessment directly probes whether generative models respond consistently and appropriately to changes in field-conditioning parameters. Combined, these five metrics provide a practical evaluation of controllability, field-awareness, and anatomically grounded structural preservation in arbitrary field-to-field MRI synthesis.

Ranking method(s)

a) Describe the method used to compute a performance rank for all submitted algorithms based on the generated metric results on the test cases. Typically the text will describe how results obtained per case and metric are aggregated to arrive at a final score/ranking. Ideally, provide the ranking scheme as a concrete pseudo code.

1. Individual Metric Ranking: For each metric (nRMSE, SSIM, LPIPS,mean Dice overlap, and mean normalized volume consistency), the metric is first computed for every test case and then averaged across all test cases. Lower values are preferred for nRMSE and LPIPS, while higher values are preferred for SSIM,Dice overlap, and normalized volume consistency.
Each algorithm is ranked independently for each metric based on its mean performance.

2. Overall Ranking: The overall ranking is obtained by summing the per-metric ranks for each algorithm. Algorithms with the lowest composite scores achieve the highest final rank. This five-metric rank-summation approach ensures that differences in metric scales do not influence the overall evaluation and provides a balanced assessment across numerical fidelity, perceptual realism, controllability, and anatomically grounded structural preservation. This procedure yields a stable and interpretable ranking that reflects performance across complementary dimensions of image quality.

b) Describe the method(s) used to manage submissions with missing results on test cases.

Participating teams are required to submit Docker containers and process all the cases in the test set on our server. For the cases without valid output, we will assign it to the lowest value of each metric.

c) Justify why the described ranking scheme(s) was/were used.

The composite rank-summation scheme is chosen to ensure fair and balanced evaluation across multiple complementary aspects of image quality. nRMSE measures numerical agreement, SSIM captures structural fidelity, LPIPS reflects perceptual realism, while Dice overlap and normalized volume consistency explicitly assess regional deep gray matter preservation. By ranking algorithms separately for each metric and combining ranks rather than raw metric values, the method avoids biases arising from differences in metric scale or dynamic range. This multi-metric ranking approach provides a robust and anatomically grounded assessment of algorithms designed for controllable and field-aware MRI generation. In addition, the top-ranked submissions will undergo blinded visual inspection by experienced neuroradiologists to identify hallucinated anatomy or anatomically implausible alterations, serving as a qualitative safeguard beyond automated metrics.

Statistical analyses

Provide an overview of the statistical approaches used in the scope of the challenge analysis. Details can be provided in the parameters below. For each parameter, justify why the described statistical method(s) was/were used and, if necessary, add a description of any method used to assess whether the data met the assumptions required for the particular statistical approach.

The statistical analysis framework of MRIxFields2026 is designed to ensure robust, unbiased, and clinically meaningful evaluation of cross-field MRI generation algorithms. Given the heterogeneous nature of the dataset—spanning magnetic field strengths from 0.1T to 7T, multiple scanners, and both paired and unpaired acquisitions—the challenge adopts case-wise, distribution-aware, and rank-robust statistical methods. Performance is primarily assessed on held-out, unseen test cases, with all metrics computed at the individual-case level and subsequently aggregated to support population-level inference. Where applicable, non-parametric statistics are preferred to avoid assumptions of normality, and all analyses are conducted under a unified evaluation protocol to ensure comparability across tasks and teams.

Provide a description of how the precision of the performance estimates of individual algorithms is assessed (e.g. confidence interval of the mean on the test set computed using percentile bootstrap, confidence interval of the accuracy on the test set computed using percentile bootstrap).

The precision of performance estimates for each submitted algorithm is quantified using bootstrap-based confidence intervals. Specifically, for each evaluation metric, percentile bootstrap resampling (e.g., 1,000 resamples with replacement at the case level) is applied to the test set to estimate 95% confidence intervals of the mean performance. This approach is well suited to the challenge setting, as it does not rely on distributional assumptions and is robust to skewed or heavy-tailed metric distributions that commonly arise in cross-domain MRI evaluation. Confidence intervals are reported alongside point estimates to reflect uncertainty and to enable fair comparison between methods.

Provide a description of how variability of the performance of individual algorithms across tests cases is assessed (e.g. SD across test cases, IQR, graphs, reporting outliers...).

To characterize how algorithm performance varies across individual test cases, case-wise variability analyses are performed. For each method and metric, variability is summarized using the standard deviation (SD) and interquartile range (IQR) across test cases, providing complementary views of dispersion and robustness. In addition, case-level performance distributions (e.g., boxplots or violin plots) are used to visualize heterogeneity, identify outliers, and highlight failure modes under specific field strengths or acquisition conditions.

Provide a description of how variability of rankings is assessed.

To evaluate the stability of algorithm rankings, variability across test cases is quantified using standard measures such as the standard deviation and interquartile range of per-metric performance. Pairwise rank correlations are computed to assess agreement between rankings derived from nRMSE, SSIM, and LPIPS. These analyses ensure that the final composite ranking is robust to minor fluctuations in individual metric values.

Provide a description of statistical tests that are used to assess whether the differences in performance between algorithms are statistically significant.

To assess whether observed performance differences between algorithms are statistically significant, paired, non-parametric hypothesis tests are employed. For pairwise comparisons between methods on the same test cases, the Wilcoxon signed-rank test is used, as it accounts for within-case pairing and does not assume normality. When comparing more than two algorithms simultaneously, a Friedman test is applied, followed by appropriate post-hoc pairwise comparisons with multiple-comparison correction (e.g., Holm–Bonferroni adjustment).

Provide a description of the missing data handling.

All test cases are verified for completeness prior to release, and missing data are not expected. If a submission fails to produce an output for a specific case, that case will be assigned the lowest possible metric score as defined in the ranking procedure, ensuring consistent treatment across teams.

Indicate any software product that is used for all data analysis methods.

All statistical analyses and visualizations are performed using Python-based tools, including NumPy, SciPy, and Matplotlib. These libraries support rank correlation analysis, variability assessment, and aggregate metric evaluation needed for the challenge.

Further analyses

Present further analyses to be performed (if applicable), e.g. related to

- combining algorithms via ensembling,
- inter-algorithm variability,
- common problems/biases of the submitted methods, or
- ranking variability.

N/A

ADDITIONAL POINTS

References

Please include any reference important for the challenge design, for example publications on the data, the annotation process or the chosen metrics as well as DOIs referring to data or code.

Reference from the organizers:
[1] Dai, Y. et al. Leveraging Deep Learning to Enhance MRI for Brain Disorders. 2025.02.10.25321126 Preprint at https://doi.org/10.1101/2025.02.10.25321126 (2025).
[2] Dai, Y., Wang, C. & Wang, H. Deep compressed sensing MRI via a gradient-enhanced fusion model. Med Phys 50, 1390–1405 (2023).
[3] Wang, F. et al. Multiple B-Value Model-Based Residual Network (MORN) for Accelerated High-Resolution Diffusion-Weighted Imaging. IEEE J Biomed Health Inform 26, 4575–4586 (2022).
[4] Wang, C. et al. Protocol for Brain Magnetic Resonance Imaging and Extraction of Imaging-Derived Phenotypes from the China Phenobank Project. Phenomics 3, 642–656 (2023).
[5] Tang, W. et al. Aleatoric-Uncertainty-Aware Maximum Intensity Projection-Based GAN for 7T-Like Generation From 3T TOF-MRA. IEEE Journal of Biomedical and Health Informatics 29, 6664–6677 (2025).
[6] Wang, Z. et al. One for multiple: Physics-informed synthetic data boosts generalizable deep learning for fast MRI reconstruction. Med Image Anal 103, 103616 (2025).

References from others:
[7] Obusez, E. C. et al. 7T MR of intracranial pathology: Preliminary observations and comparisons to 3T and 1.5T. Neuroimage 168, 459–476 (2018).
[8] Maranzano, J. et al. Comparison of Multiple Sclerosis Cortical Lesion Types Detected by Multicontrast 3T and 7T MRI. AJNR Am J Neuroradiol 40, 1162–1169 (2019).
[9] Cosottini, M. et al. Comparison of 3T and 7T susceptibility-weighted angiography of the substantia nigra in diagnosing Parkinson disease. AJNR Am J Neuroradiol 36, 461–466 (2015).
[10] Li, S. et al. Synthetizing SWI from 3T to 7T by generative diffusion network for deep medullary veins visualization. Neuroimage 320, 121475 (2025).
[11] Lv, J. et al. Transfer learning enhanced generative adversarial networks for multi-channel MRI reconstruction. Comput Biol Med 134, 104504 (2021).
[12] Qu, L., Zhang, Y., Wang, S., Yap, P.-T. & Shen, D. Synthesized 7T MRI from 3T MRI via deep learning in spatial and wavelet domains. Med Image Anal 62, 101663 (2020).
[13] Duan, C. et al. Synthesized 7T MPRAGE From 3T MPRAGE Using Generative Adversarial Network and Validation in Clinical Brain Imaging: A Feasibility Study. J Magn Reson Imaging 59, 1620–1629 (2024).
[14] Eidex, Z. et al. High-resolution 3T to 7T ADC map synthesis with a hybrid CNN-transformer model. Med Phys

51, 4380–4388 (2024).
[15] Bahrami, K., Rekik, I., Shi, F. & Shen, D. Joint Reconstruction and Segmentation of 7T-like MR Images from 3T MRI Based on Cascaded Convolutional Neural Networks. Med Image Comput Comput Assist Interv 10433, 764–772 (2017).
[16] Sun, Y., Wang, L., Li, G., Lin, W. & Wang, L. A foundation model for enhancing magnetic resonance images and downstream segmentation, registration and diagnostic tasks. Nat Biomed Eng 9, 521–538 (2025).
[17] Billot, B. et al. SynthSeg: Segmentation of brain MRI scans of any contrast and resolution without retraining[J]. Medical image analysis, 86: 102789(2023).

Further comments

Further comments from the organizers.

This challenge addresses a critical gap in current MRI research: the lack of standardized benchmarks for AI models operating across heterogeneous field strengths, from ultra-low-field to ultrahigh-field systems. By focusing on high-field-equivalent synthesis, low-field enhancement, and controllable field-to-field generation, the challenge directly responds to pressing clinical and technical needs in multi-center harmonization and global MRI accessibility. We have secured diverse multi-field datasets, established robust evaluation protocols, and engaged a highly interested research community. The organizing team is fully prepared for data release, platform operation, and post-challenge dissemination, ensuring strong scientific impact and meaningful contributions to field-aware MRI development.

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